Tournaments not inducible by five voters

In loving memory of Adam Paton.

Leonid Chindelevitch[a] **and Ararat Harutyunyan**[b]

[a] MRC Centre for Global Infectious Disease Analysis, School of Public Health, Imperial College London, 90 Wood Lane, London W12 0BZ, United Kingdom. Email: `lchindel@ic.ac.uk`. ORCID: 0000-0002-6619-6013.

[b] University of Paris-Dauphine, PSL University, CNRS UMR7243, LAMSADE, Place du Maréchal de Lattre de Tassigny, 75775 Paris Cedex 16, France. Email: `ararat.harutyunyan@lamsade.dauphine.fr`.

Corresponding author: Leonid Chindelevitch (`lchindel@ic.ac.uk`).

Abstract

How many voters are needed before their pairwise majorities can produce every tournament? For odd k, let $N(k)$ be the least number of vertices on which some tournament arises from no k voters. Only $N(3) = 8$ is known (Shepardson and Tovey 2009; Eggermont et al. 2013). For five voters the best bounds were $11 \leq N(5)$ (Bachmeier et al. 2019) and $N(5) \leq 41$ (Eggermont et al. 2013, p. 271; Bachmeier et al. 2019). We show $13 \leq N(5) \leq 22$. The upper bound comes from nine tournaments on 22 vertices that five voters cannot produce: eight one-vertex deletions of doubly regular tournaments on 23 vertices, including every deletion of one of them, and one arc reversal. The Paley tournament on 23 vertices, which Bachmeier et al. (2019) left open after six weeks of SAT solving, is beyond five voters, and so is every other doubly regular tournament on 23 vertices. The lower bound comes from an analysis of all tournaments on 12 vertices. An argument that searches no profiles shows that the Paley tournament on 43 vertices is also beyond five voters. Its predictability, the largest share of the voters that a profile can give to every majority at once, exceeds the threshold 3/5 by only 2/1505, so this result refutes the conjecture that a predictability above the threshold lets five voters produce a tournament. Our method is a search that places one vertex at a time and exploits the tournament's symmetries. We certify four of the negative results with machine-checkable proofs, and a public repository reproduces every result.

Keywords: majority voting; tournaments; McGarvey's theorem; Paley tournaments; constraint search; SAT

1. Introduction

Let T be a *tournament* on a vertex set V with $|V| = n$: a digraph in which every unordered pair $\{i, j\}$ carries exactly one of the arcs $i \to j$ and $j \to i$. A *profile* of k *voters* is a list of k linear orders $\pi_1, \ldots, \pi_k$ of V, and its *majority tournament* has the arc $i \to j$ whenever $|\{v : i \prec_{\pi_v} j\}| > k/2$; for odd k this decides every pair. We say that T is *k-inducible* if it is the majority tournament of some profile of k voters, which we then call a *witness* for T. McGarvey (1953) proved that every tournament is k-inducible for some k, and counting profiles shows that for fixed k almost all large tournaments are not. For odd k we write $N(k)$ for the least order of a tournament that is not k-inducible.

Only $N(3) = 8$ is known (Shepardson and Tovey 2009; Eggermont, Hurkens and Woeginger 2013). For $k = 5$ the bounds were $11 \leq N(5) \leq 41$ (Bachmeier et al. 2019; Eggermont et al. 2013, p. 271), the upper one by counting, and the smallest explicit tournament known not to be 5-inducible had over 6×10^8 vertices (Fidler 2011; Bachmeier et al. 2019). We show that $13 \leq N(5) \leq 22$; Section 1.3 lists our results, and Figure 1 shows one of the eight tournaments on 22 vertices that give the upper bound.

What these *obstructions* (tournaments that are not 5-inducible) share is balance rather than symmetry: none of the 37 doubly regular tournaments on 23 vertices, in which every vertex has out-degree 11 and every two vertices the same number of common out-neighbours, is 5-inducible, although 32 of them have a trivial automorphism group (Section 6.4).

The *support* $c(e)$ of an arc $e : i \to j$ is the number of voters ranking i above j, and its *margin* is $2c(e) - k$, so for $k = 5$ every margin in a witness is 1, 3 or 5. A witness has *margin at most M* if every arc does, and *unit margin* if every margin is 1. Our companion paper (Chindelevitch and Harutyunyan 2026), whose definitions

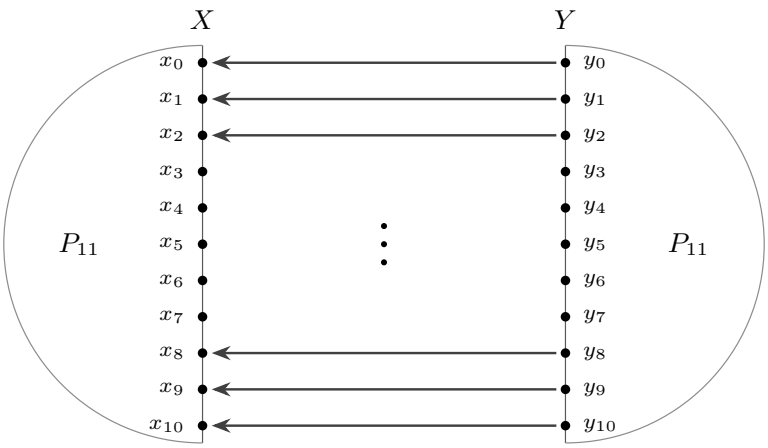


Figure 1: The tournament $\Delta_1 - v_{10}$ (notation of Section 1.3), which is not 5-inducible. Its 22 vertices split into $X = \{x_0, \ldots, x_{10}\}$ and $Y = \{y_0, \ldots, y_{10}\}$, with indices taken modulo 11, and each half induces a copy of P_{11}: $x_i \to x_j$ exactly when $j - i$ is a non-zero square modulo 11, that is, lies in $\mathrm{QR} = \{1, 3, 4, 5, 9\}$, and $y_i \to y_j$ exactly when $j - i$ is a non-square. Between the halves, $x_i \to y_j$ when $j - i$ is a non-square, and $y_j \to x_i$ otherwise, that is, when $j - i$ lies in $\mathrm{QR} \cup \{0\}$. In particular every rung $y_i \to x_i$ of the ladder points from Y to X; only six rungs are drawn. Each x_i therefore has out-degree 10 and each y_j has out-degree 11. Adding 1 to every index, and multiplying every index by an element of QR, are automorphisms; together they generate the full automorphism group, of order 55.

we use, conjectures that this margin hierarchy is strict; for $k = 5$, P_{19} is 5-inducible with margin at most 3 but not with unit margin (Section 6.2). Reversing every voter reverses every arc and changes no support, so T and its converse $\overline{T}$ are k-inducible for the same k and at the same margins; we count a family enumerated in full up to isomorphism, and a family that our searches generated up to isomorphism and converse.

The *predictability* $\alpha^*(T)$ (Shepardson and Tovey 2009) is the largest $\alpha > 1/2$ such that some profile, on any number t of voters, supports every arc of T with at least αt of them. A k-voter witness supports every arc with more than $k/2$ voters, so k-inducibility forces $\alpha^*(T) \geq (k+1)/2k$, which is $3/5$ at $k = 5$; the converse, implicit in their work, we call the *threshold conjecture*. Our companion paper refutes it at $k = 3$, where it fails only at the threshold; at $k = 5$ it fails strictly above it, for P_{23} with $\alpha^* = 7/11$ (Section 6.1) and for P_{43}, where α^* exceeds $3/5$ by only $2/1505$ (Section 4).

1.1 Special properties of Paley tournaments

The Paley tournament P_q, for a prime $q \equiv 3 \pmod 4$, has vertex set $\mathbb{F}_q$ and an arc from i to j whenever $j - i$ is a nonzero square, or *quadratic residue*; we write QR for the set of these, so that $i \to j$ exactly when $j - i \in \mathrm{QR}$. As everywhere in this paper, vertices are numbered from 1, vertex $i + 1$ being the field element i. It is *doubly regular*: every vertex has out-degree $(q-1)/2$ and every ordered pair of vertices has exactly $(q-3)/4$ common out-neighbours. Its automorphism group $\mathrm{Aut}(P_q)$ (Section 2) has order $q(q-1)/2$. This makes P_q *arc-transitive*, meaning that for every pair of arcs e, e' of P_q there is an automorphism of P_q that takes e into e'. It is also *self-converse*, meaning that it is isomorphic to its own converse; the relevant mapping is multiplication by -1, which is easily checked to not be an automorphism when $q \equiv 3 \pmod 4$.

1.2 Neither SAT nor the slack argument settles Paley(23)

Paley tournaments are natural candidates for non-inducibility because they are far from being transitive, and because their symmetry makes an otherwise hopeless search merely very hard. Bachmeier et al. (2019) call them *quadratic residue tournaments*, with the converse orientation, which changes no result here. Their SAT study shows that P_3 and P_7 are 3-inducible and that P_{11} and P_{19} are 5-inducible but not 3-inducible, and then stops at P_{23}, on which their solver did not terminate within six weeks.

Nor does the argument by which we settle P_{43} (Section 4), which searches no profiles. It rests on the *slack*, the amount by which five voters can collectively fall short of the most arcs one order can agree with; the slack is proportional to $\alpha^* - 3/5$, and it is 6 at $q = 43$ but 46 at $q = 23$ (Table 2), which leaves the voters of a candidate profile for P_{23} far too much room.

1.3 Contributions

1. **There are non-5-inducible tournaments on 22 vertices, so $N(5) \leq 22$** (Section 6.5). We write Δ_i for the i-th of the 37 doubly regular tournaments on 23 vertices in the catalogue of McKay (2026), and $\Delta_i - v_j$ for Δ_i with its vertex v_j deleted. Eight such tournaments, pairwise non-isomorphic even up to converse, are not 5-inducible and are vertex-critical; three are the one-vertex deletions of Δ_1, and reversing an arc of one of them, $\Delta_1 - v_{10}$, gives a ninth obstruction.
2. **P_{23} is not 5-inducible** (Section 6.1). The SAT solver of Bachmeier et al. (2019) did not terminate on it within six weeks; our search settles it in twenty-two hours on a laptop. We also decide P_{31}.
3. **Counting gives $N(5) \leq 38$, and P_{43} is not 5-inducible** (Sections 3 and 4). The counting bounds of Bachmeier et al. (2019) improve for every odd $k \leq 21$, and P_{43}, whose predictability exceeds 3/5 by only 2/1505, refutes the threshold conjecture at $k = 5$ strictly above the threshold.
4. **P_{19} is 5-inducible, but not with unit margin** (Section 6.2). At $k = 3$ such tournaments first appear at order 9, after $N(3) = 8$ (see our companion paper); at $k = 5$ they appear by order 19, before $N(5)$ unless $N(5) \leq 19$, which the search of Appendix C makes unlikely.
5. **P_{19} and P_{23} are arc-critical, and P_{31} and P_{43} are not vertex-critical** (Sections 6.1 and 6.3). Reversing an arc of P_{19} restores 5-inducibility at unit margin and reversing an arc of P_{23} restores it unrestricted, while P_{31} and P_{43} minus a vertex are still not 5-inducible.
6. **Four negative results are formally certified** (Appendix B): P_{19} at unit margin, and P_{23}, $\Delta_1 - v_{10}$ and $\Delta_6 - v_5$. Each subproblem (*cube*) is refuted by the SAT solver CaDiCaL with a machine-checkable *LRAT* proof (Section 1.4), rechecked by a separate program.
7. **Every tournament on 12 vertices is 5-inducible, even at unit margin** (Appendix A), by settling all 2,048 one-vertex extensions of each of the $\approx 9.0 \times 10^8$ tournaments on 11 vertices.

1.4 Related work

This paper builds on two others. Our companion paper (Chindelevitch and Harutyunyan 2026) settles the three-voter questions from which this one starts, and we use its definitions. Bachmeier et al. (2019) is the prior art for both the *bounds* we improve and the *method*: besides the bounds above, it brought SAT to bear on k-majority digraphs, characterized the 3-majority case building on Dushnik and Miller (1941), and established hardness results for several voting rules with a constant number of voters. Without a bound on k, every tournament on n vertices is k-inducible with $k = O(n/\log n)$ (Erdős and Moser 1964).

The value of $N(3)$ is settled. Shepardson and Tovey (2009) introduced the predictability $\alpha^*(T)$ defined above. A 3-voter profile is in particular a 2/3-supermajority representation, so $\alpha^*(T) < 2/3$ rules out 3-inducibility; their 8-vertex tournament has $\alpha^* = 13/20$, which gives $N(3) \leq 8$. They also prove that every tournament on at most 7 vertices has $\alpha^* \in \{2/3, 1\}$, and Eggermont, Hurkens and Woeginger (2013) then show by direct computation using an integer linear programming (ILP) formulation that every tournament on 7 vertices is 3-inducible, and every tournament on 8 or 9 vertices is 5-inducible, yielding $N(3) = 8$ and $N(5) \geq 10$.

So at $k = 3$ predictability decides exactly at $N(3)$; at $k = 5$ it does not, as P_{23}, P_{31} and P_{43} show (Table 2).

Our certified computation (Appendix B) is standard. It splits the problem into independent subproblems, using the *cube-and-conquer* method (Heule et al. 2012): a SAT instance is split by fixing a few variables at a time, giving many partial assignments called *cubes*, and each cube is handed to the solver as a problem of its own. We adopt the standard of evidence set by the Boolean Pythagorean triples proof (Heule et al. 2016), later re-checked with a formally verified checker (Cruz-Filipe et al. 2019), in which every subproblem emits a machine-checkable refutation that an independent checker validates. Our refutations are in the LRAT format

(Cruz-Filipe et al. 2017) (Appendix B.1), a format in which the solver records each clause it derives so that an independent checker can replay and confirm the entire derivation.

The rest of this paper is organized as follows. Section 2 sets out the notation. Section 3 bounds $N(k)$ by counting, and Section 4 proves that P_{43} is not 5-inducible by an argument that searches no profiles. Section 5 describes the search: its symmetry reductions, the algorithm and what makes it fast. Section 6 presents the results of the search: the Paley tournaments (Sections 6.1 to 6.3), the other structured families (Section 6.4), and the eight 22-vertex obstructions that give $N(5) \leq 22$ (Section 6.5). Section 7 states what remains open and describes the reproducibility material.

2. Preliminaries

In this section we fix the notation used throughout.

Throughout, T is a tournament on $V = \{1, \ldots, n\}$ and $k = 5$ unless stated otherwise. We write $N^+(v)$ and $N^-(v)$ for the out- and in-neighbourhoods of v. A tournament is *regular* if $|N^+(v)| = |N^-(v)|$ for each $v \in V$; this can only happen for n odd. For even n it is *near-regular* if every out-degree is $n/2 - 1$ or $n/2$; since the out-degrees sum to $\binom{n}{2}$, exactly $n/2$ vertices then have each of the two. Following our companion paper, D_n, R_n and S_n denote the numbers of isomorphism classes of, respectively, all tournaments, regular tournaments and self-converse tournaments on n vertices, which are OEIS A000568, A096368 and A002785 (OEIS Foundation Inc. 2026a, 2026c, 2026b). For any $S \subseteq V$, the sub-tournament of T induced by S, written $T[S]$, is the tournament obtained by restricting T to the vertices in S and the arcs between those vertices.

We call a tournament T *vertex-critical* for k-inducibility at margin M if T is not k-inducible at margin M, but $T - v$ is k-inducible at margin M for every vertex v. Similarly, we call T *arc-critical* for such a property if T has it, but any T' obtained from T by reversing an arc does not.

An *automorphism* of a tournament T is a permutation of its vertices carrying every arc to an arc; these form a group $\mathrm{Aut}(T)$ under composition. The identity is always an automorphism; any other automorphism is *non-trivial*, and T is *rigid* when it has none. For a permutation group Γ on a set X, the *orbit* of x is $\Gamma x = \{\gamma(x) : \gamma \in \Gamma\}$, and the orbits partition X. The group is *transitive* on X when there is a single orbit, that is, when for any $x, x' \in X$ there is a $g \in \Gamma$ with $g(x) = x'$. We use orbits of $\mathrm{Aut}(T)$ on the vertices, on the arcs, and on the *ordered pairs* of distinct vertices, where γ sends (u, v) to $(\gamma(u), \gamma(v))$. An ordered pair is either an arc or the reverse of one, so that third action has at least two orbits.

The next section uses this notation to bound $N(k)$ by counting.

3. Counting bounds on $N(k)$

In this section we bound $N(k)$ by counting, without exhibiting a tournament: some tournament on 38 vertices is not 5-inducible, and the counting bounds of Bachmeier et al. (2019) improve for every odd $k \leq 21$.

Comparing counts of profiles and tournaments gives $N(5) \leq 41$ (Eggermont et al. 2013, p. 271; Bachmeier et al. 2019). Restricting the count to *regular* tournaments tightens this to $N(5) \leq 39$, and to *near-regular* tournaments to $N(5) \leq 38$, because four voters determine a tournament of prescribed out-degrees.

3.1 Four voters determine a tournament with given out-degrees

The fifth voter of a profile is constrained only on the pairs that the other four split evenly, and there a prescribed out-degree sequence leaves it no choice.

Proposition 3.1 (completion uniqueness). *Fix odd k, linear orders $\pi_1, \ldots, \pi_{k-1}$ on $[n]$ and a sequence $d\colon [n] \to \mathbb{Z}_{\geq 0}$. At most one tournament with out-degree sequence d is the majority tournament of $\pi_1, \ldots, \pi_{k-1}$ and some k-th order, and it can be found in $O(n^2)$ time.*

Proof. Pairs that $\pi_1, \ldots, \pi_{k-1}$ split with margin at least 2 are decided whatever π_k does; the evenly split pairs form a graph D on which the majority follows π_k. The tournament is therefore determined by the orientation R of D that π_k induces, which is acyclic, and every acyclic orientation of D arises from some

π_k. The sequence d fixes the out-degree $b(x)$ of each x in R: $d(x)$ minus the number of decided arcs leaving x. An acyclic orientation has a sink, a vertex with $b(x) = 0$, and such a vertex must have all its remaining edges oriented into it. Repeatedly deleting a vertex with $b(x) = 0$, orienting its remaining edges into it and decrementing its neighbours' budgets therefore either fails, when no such vertex exists or a budget turns negative, and then no completion exists, or orients every edge in the only possible way; the reverse deletion order is then a valid π_k (Kahn 1962). Each vertex and edge is handled once, in $O(n^2)$ time. ■

3.2 Counting near-regular tournaments gives $N(5) \le 38$

For even n there are no regular tournaments, but the near-regular ones (Section 2) substitute for them: deleting a vertex v from a regular tournament on $n+1$ vertices leaves a near-regular tournament on n, and each near-regular tournament arises from exactly one regular tournament in this way, by re-adding v beating the $n/2$ vertices of out-degree $n/2-1$. Hence

$$\#\{\text{near-regular tournaments on } n\} \;=\; R^L_{n+1}, \tag{3.1}$$

where R^L_ν is the number of *labelled* regular tournaments on ν vertices, the counting taking place in the labelled world, unlike the isomorphism count R_n.

Theorem 3.2. *Some near-regular tournament on 38 vertices is not 5-inducible; hence $N(5) \le 38$.*

Proof. Deleting a voter from a witness shows, by Proposition 3.1, that a 5-inducible tournament with a prescribed out-degree sequence is determined by a 4-multiset of orders. The near-regular sequences on even n number $\binom{n}{n/2}$, so

$$\log_2 \#\{\text{5-inducible near-regular on } n\} \;\le\; \log_2\Big[\binom{n}{n/2}\binom{n!+3}{4}\Big] \;<\; 624.7 \quad (n = 38).$$

By (3.1) and Schrijver's (1983) lower bound on Eulerian orientations of K_{39}, of which the regular tournaments on 39 vertices are an instance,

$$\log_2 \#\{\text{near-regular on } 38\} \;=\; \log_2 R^L_{39} \;\ge\; \log_2\Big[2^{-\binom{39}{2}}\binom{38}{19}^{39}\Big] \;>\; 625.5,$$

which exceeds 624.7. ■

The regular count alone gives only $N(5) \le 39$: $\log_2 R^L_{39} > 625.5 > 610.8 > \log_2 \binom{39!+3}{4}$, so all but a $2^{-14.8}$ fraction of regular tournaments on 39 vertices are not 5-inducible. At $k = 3$ the same argument shows only that some regular tournaments on 17 or more vertices are not 3-inducible, far weaker than $N(3) = 8$.

3.3 The counting bound improves for every odd $k \le 21$

The argument is not specific to five voters: Proposition 3.1 holds for every odd k, and the near-regular bijection (3.1) does not depend on k. Counting both classes gives:

Theorem 3.3 (all odd k). *For odd $k \ge 3$: (a) at most $\binom{n!+k-2}{k-1}$ regular tournaments on odd $[n]$ are k-inducible, so a non-k-inducible regular tournament exists once $R^L_n > \binom{n!+k-2}{k-1}$; (b) at most $\binom{n}{n/2}\binom{n!+k-2}{k-1}$ near-regular tournaments on even $[n]$ are k-inducible, so a non-k-inducible near-regular tournament exists once $R^L_{n+1} > \binom{n}{n/2}\binom{n!+k-2}{k-1}$, using in both cases Schrijver's bound $R^L_\nu \ge 2^{-\binom{\nu}{2}}\binom{\nu-1}{(\nu-1)/2}^{\nu}$ (odd ν).*

Together with the exact multiset count of profiles, this extends Table 1 of Bachmeier et al. (2019); Table 1 lists both. Their counting lemma bounds the majoritarian expressiveness $n^{\mathcal{T}}(k)$, the largest n such that every n-vertex tournament is k-inducible, via the estimate $\binom{n!+k-1}{k} \le (2\cdot n!)^k/k!$. Using the exact multiset count in place of their estimate, which uses the bound of Fiol (1992) (the "exact" row), already sharpens three entries (the $k = 5$ case noted in a footnote of Bachmeier et al. 2019); the regular and near-regular bounds of Theorem 3.3 then sharpen all ten. The 10 best bounds are equally split between *regular* tournaments of odd size and *near-regular* tournaments of even size. The smallest non-k-inducible tournament has $N(k) = n^{\mathcal{T}}(k) + 1$ vertices, so that $n^{\mathcal{T}}(5) \le 37$ is exactly the bound $N(5) \le 38$ of Theorem 3.2.

Table 1: Upper bounds on the majoritarian expressiveness $n^{\mathcal{T}}(k)$, the largest n such that every n-vertex tournament is k-inducible. Columns are indexed by the odd voter count k; bold marks a strict improvement on Table 1 of Bachmeier et al. (2019).

	3	5	7	9	11	13	15	17	19	21
Bachmeier et al. (2019)	18	41	66	93	122	152	183	216	249	282
exact	18	**40**	**65**	93	122	152	183	**215**	249	282
Thm 3.3	**16**	**37**	**62**	**89**	**117**	**147**	**178**	**210**	**243**	**276**

Each threshold is sharp: the count wins at order $n^{\mathcal{T}}(k)+1$ and fails one vertex lower.

These bounds show that a non-5-inducible tournament exists on 38 vertices, but they do not exhibit one. The next section gives an explicit example on 43 vertices.

4. Paley(43) is not 5-inducible

P_{43} is not 5-inducible, although its predictability exceeds the threshold 3/5. Section 3 showed that a non-5-inducible tournament exists on 38 vertices; in this section we prove that P_{43} is one, by an argument that searches no profiles. Among the Paley tournaments of Table 2, P_{43} is the only one that the margin of its predictability over 3/5 settles on its own: that margin is just 2/1505, and it leaves five voters too little room. P_{23}, with $\alpha^* = 7/11$, refutes the threshold conjecture further above the threshold (Section 6.1), but only through the search. We write $P = P_{43}$ throughout the section.

Theorem 4.1. *$P = P_{43}$ satisfies $\alpha^*(P) = \mathrm{MAS}/C = 543/903 = 181/301 > 3/5$, yet P is not 5-inducible. Hence $\alpha^* > \frac{k+1}{2k}$ does not imply k-inducibility, and $N(5) \le 43$ with an explicit tournament.*

Here $\mathrm{MAS}(T)$ is the maximum number of arcs of T that a single linear order agrees with. An arc $u \to v$ is *forward* in an order π if π ranks u above v and *backward* otherwise, and $\mathrm{fwd}_T(\pi)$ is the set of forward arcs. The identity $\alpha^* = \mathrm{MAS}/C$ holds because P is arc-transitive, that is, $\mathrm{Aut}(P)$ acts transitively on its arcs. No order has more than MAS forward arcs, so in any profile the average support of an arc is at most a fraction MAS/C of the voters, and the smallest support is no larger. Conversely, the profile formed by the images under $\mathrm{Aut}(P)$ of one order with MAS forward arcs gives every arc exactly that fraction (the symmetrization lemma of our companion paper). Throughout, $q = 43$, $C = \binom{q}{2} = 903$, the number of cyclic triangles is $N_\Delta = (q^3 - q)/24 = 3311$, and $|\mathrm{Aut}(P)| = q(q-1)/2 = 903$.

4.1 The slack forces two near-optimal voters

In a witness every arc is forward for at least three of the five voters, so the voters have at least $3C$ forward arcs between them, while none has more than MAS. The *slack*

$$\mathrm{slack}_5(T) \;=\; 5\,\mathrm{MAS}(T) - 3C$$

therefore bounds how far the voters can collectively fall short of the maximum. For an arc-transitive T it equals $5C\,(\alpha^*(T) - 3/5)$, so it grows in proportion to C whenever the predictability stands off from the threshold. A subset dynamic program modulo $\mathrm{Aut}(P)$ with a meet-in-the-middle step (Appendix D) gives $\mathrm{MAS}(P) = 543$, so $\mathrm{slack}_5(P) = 2715 - 2709 = 6$, the smallest value in Table 2; the others are at least 7.

For an inducing profile $(\pi_1, \dots, \pi_5)$, let $f_i = |\mathrm{fwd}_P(\pi_i)|$ and relabel the voters so that $f_1 \ge f_2 \ge \cdots \ge f_5$. Then $\sum_{i=1}^5 f_i \ge 3C = 2709$ and $f_1 \le 543$, so the other four sum to at least 2166, and $4f_2 \ge 2166$, i.e. $f_2 \ge 542$. The *level* of an order π is $\delta(\pi) = \mathrm{MAS} - |\mathrm{fwd}_P(\pi)|$, so **at least two voters lie at level** ≤ 1, and the five levels sum to at most $\mathrm{slack}_5(P) = 6$.

A cyclic triangle is *double-backed* by an order that ranks two of its three arcs backward, and the *double-back set* $\mathrm{DB}(\pi)$ of π is the set of cyclic triangles it double-backs.

Lemma 4.2 (co-backing). *In a 5-voter witness, no cyclic triangle is double-backed by two voters.*

Proof. Each arc of the triangle is ranked backward by ≤ 2 of the 5 voters, so there are at most six backward incidences; each voter ranks ≥ 1 arc of a cyclic triangle backward (a linear order cannot agree with all three arcs of a cycle); if d voters double-back, incidences $= 5 + d \leq 6$, so $d \leq 1$. ■

The five voters' double-back sets are therefore pairwise disjoint, so the two voters at level ≤ 1 have disjoint double-back sets. **If no such pair exists, P is not 5-inducible**, and this is a finite criterion.

4.2 No two near-optimal orders have disjoint double-back sets

The orders of level at most 1, which we call the *shell*, number 1,662,696,609, in 1,841,303 orbits of $\mathrm{Aut}(P)$ (Appendix E). We decide the criterion by checking every pair, which two reductions make practical:

- **A window of 24 vertices.** We fix the window $\mathcal{W} = \{0, 1, \ldots, 23\}$ and its 538 cyclic triangles. The restriction of $\mathrm{DB}(\pi)$ to these triangles depends only on how π ranks $\mathcal{W}$, so the shell has only 4,709,640 distinct restrictions, and two orders with disjoint double-back sets have disjoint restrictions; only the 5,092,111 disjoint pairs of restrictions need the exact check.
- **Orbit representatives on one side.** $\mathrm{DB}(\sigma\pi) = \sigma(\mathrm{DB}(\pi))$ for $\sigma \in \mathrm{Aut}(P)$, where $\sigma\pi$ ranks $\sigma(v)$ where π ranks v, because σ maps forward arcs of π to forward arcs of $\sigma\pi$. Overlap sizes are thus Aut-invariant, so testing one representative per orbit against the shell suffices, saving a factor of 903.

The exact check compared all 4,376,325,129 pairs of an orbit representative and a shell order whose restrictions are disjoint, and **none has disjoint double-back sets**; the smallest overlap is 68 triangles. With the co-backing lemma, this proves Theorem 4.1. ■

Table 2: The Paley family: size, MAS, predictability, slack and status. For $q = 47$ the entries are lower bounds, from an order with 658 forward arcs.

q	C	MAS	α^*	$\approx$	slack_5	5-inducible?
7	21	14	2/3	0.6667	7	yes; 3 voters suffice
11	55	35	7/11	0.6364	10	yes; 5 voters needed
19	171	107	107/171	0.6257	22	yes; 5 voters needed
23	253	161	7/11	0.6364	46	no (Thm 6.2)
31	465	285	19/31	0.6129	30	no (Sec. 6.1)
43	903	543	181/301	0.6013	**6**	**no (Thm 4.1)**
47	1081	≥ 658	$\geq 14/23$	0.6087	≥ 47	open

The table shows that α^* is not monotone in q: it coincides at $q = 11$ and $q = 23$, and is larger at $q = 47$ than at $q = 43$, so it cannot be extrapolated. The slack is small only at $q = 43$, so the proof above exploits a numerical coincidence. P_{11} and P_{19} are 5-inducible, so non-5-inducibility is not generic to the family; the slack argument does not reach P_{23} or P_{31}, which the search of Section 6 decides, and P_{47} remains open.

Whether $\alpha^* > \frac{k+1}{2k}$ implies k-inducibility for odd $k \geq 7$ is open. Adding an order and its reverse changes no margin, so k-inducibility implies $(k+2)$-inducibility, and a counterexample at $k = 7$ must be non-5-inducible. P_{43} meets the $k = 7$ threshold ($181/301 > 4/7$), but the slack arithmetic and the co-backing lemma both fail at $k = 7$. Theorem 3.3 guarantees non-7-inducible tournaments on 63 vertices, without exhibiting one.

Before this work, the smallest explicit non-5-inducible tournament had over 6×10^8 vertices, from a theorem of Fidler (2011) (Bachmeier et al. 2019). The slack argument reaches no further than P_{43}, so the next section describes the search, which settles P_{23} and P_{31} and finds the obstructions on 22 vertices.

5. The search

In this section we describe the search that decides the instances of Section 6: the symmetry reductions it relies on (Section 5.1), the algorithm (Section 5.2) and what makes it fast (Section 5.3). Appendix F proves the reductions and explains why integer programming and SAT alone fall short here.

5.1 Base states and symmetry reductions

Every search starts from a small set B of vertices, the *base*, and a *base state* of B: k linear orders of B whose majorities reproduce $T[B]$ at the margin being asked for, counted up to relabelling the voters. Each base state is a self-contained subproblem, namely whether the remaining $n - |B|$ vertices can be inserted so that every arc of T gets the right support, and T is k-inducible exactly when some base state extends to a witness. The number of base states depends on $|B|$, on $T[B]$ and on the margin, but not on n: the five-vertex bases we use on the Paley tournaments have 8,031 of them in both P_{23} and P_{47}.

Two symmetries act on witnesses: the $k!$ relabellings of the voters, which form the symmetric group S_k, and $\mathrm{Aut}(T)$ acting on the vertices. A permutation σ of the voters acts on a profile $\pi = (\pi_1, \ldots, \pi_k)$ by reordering it, and a permutation h of V acts on a linear order by relabelling, so that $h\pi_j$ ranks $h(u)$ above $h(w)$ exactly when π_j ranks u above w, and on a profile voter by voter. The two actions commute, so the group $S_k \times \mathrm{Aut}(T)$ acts on profiles. It maps witnesses to witnesses: permuting voters does not change any margin, and the margin of $h(u) \to h(w)$ after relabelling by an automorphism h equals that of $u \to w$ before. A witness *extends* the base state formed by its orders restricted to B, and the *restriction* of a base state of B to $C \subseteq B$ is the multiset of its orders restricted to C. Every reduction we use is a case of the following lemma, which Appendix F.2 proves.

Lemma 5.1 (symmetry reductions). *Let T be k-inducible.*

(a) (Orbit principle.) Let G be a subgroup of $S_k \times \mathrm{Aut}(T)$ acting on a set Ω, let φ map profiles to Ω with $\varphi(g\pi) = g\,\varphi(\pi)$ for every $g \in G$, and let $R \subseteq \Omega$ meet every G-orbit. Then T has a witness π with $\varphi(\pi) \in R$.

(b) (Orbit anchoring.) If $(a_1, b_1), \ldots, (a_m, b_m)$ represent the orbits of $\mathrm{Aut}(T)$ on ordered pairs of distinct vertices, then T has a witness in which, for some i, some voter ranks a_i first and b_i second.

(c) (Vertex anchoring.) If $v_1, \ldots, v_r$ represent the vertex orbits, a voter of some witness ranks some v_i first.

(d) (Voter sorting.) For any total order on linear orders, T has a witness whose k orders are non-decreasing, and this witness can also be taken to satisfy (b) or (c).

(e) (Base-stabilizer quotient.) Let H be a subgroup of $\mathrm{Aut}(T)$ mapping $C \subseteq B$ onto itself, choose one base state of C from each H-orbit, and call a base state of B chosen if its restriction to C is chosen. Then some chosen base state of B extends to a witness. If every element of H fixes each representative of (b), or of (c), then that witness can also be taken to satisfy (b), or (c), respectively.

We call a witness satisfying (b) or (c) *anchored*. Sorting the voters (part (d)) is built into the definition of a base state and divides their number by almost 120. Anchoring discards base states outright when every representative lies in B, leaving 2,591 of the 8,031 *live* on a Paley tournament, and otherwise it prunes during the search. The quotient (e) divides the work by up to $|H|$. The two combine when H fixes every representative, so we take H to be their common stabilizer, and together they use all of $\mathrm{Aut}(T)$ (Table F.1).

5.2 The placement search

The search maintains a partial profile: a set $S \subseteq V$ of *placed* vertices together with, for each voter, a linear order of S. It begins from a base state, which places B, and inserts one unplaced vertex at a time. Inserting v means choosing, for each voter independently, a slot among the $|S| + 1$ available; the *domain* $D(v \mid S)$ is the set of k-tuples of slots consistent with every arc between v and S. A partial profile is extendable only if every unplaced vertex has a non-empty domain, and domains shrink as S grows. Algorithm 1 gives the search, and Figure F.2 (Appendix F.3) runs it on P_7 at $k = 3$. The base states are searched in parallel; the verdict is positive as soon as one yields a witness, and negative only when every live base state has been searched to exhaustion. The certified refutations of Appendix B use the same decomposition, handing each live base state to a conflict-driven clause-learning (CDCL) solver as a *cube* (the partial assignment fixing the voters' rankings of B), so the two share their symmetry reductions and differ only in what refutes a leaf.

5.3 What makes the search fast

Five choices make the search fast; the second is the one that made $n = 19$ and $n = 23$ reasonable to compute.

1. **A dynamic variable order.** At every node we insert the unplaced vertex of smallest domain (the minimum-remaining-values, or MRV, rule), breaking ties toward the vertex whose arcs to S are most imbalanced. On Paley instances a static order is not competitive, because how constrained a vertex is depends strongly on which part of the tournament has already been placed.
2. **Incremental domain refinement.** A child's domains are derived from its parent's by `REFINE`, in time proportional to the surviving tuples rather than to the whole slot space, and they live in a stack-disciplined arena indexed by (depth, vertex), so that backtracking is a pointer reset. Like for like this is worth a factor of about 110 in wall time.
3. **Symmetry reductions.** Base states are sorted and taken one per orbit of H, and the anchors are tested at every node once all representatives are placed (Section 5.1).
4. **Early termination and caching.** A per-voter agreement bound, counting the agreement between a voter and the arcs between v and S two ways, proves $D(v \mid S)$ empty in $O(k|S|)$ time without enumerating it. Base-state counts depend only on the arc pattern of $T[B]$, so they are reused across every copy of the same base class and across every q.
5. **Choosing the base.** The fewest live base states is not the fastest choice: at $q = 31$ the class with 4,007 live states beat the one with 2,537, whose states are individually slower, so we rank the twelve five-vertex classes by the live count times a sampled time per state.

The next section reports what this search decides.

6. Results

In this section we present our results, first for the Paley tournaments and the other structured families, then for the eight obstructions on 22 vertices. Table B.3 in Appendix B.2 lists what each result rests on.

The two regimes we explore are *unrestricted* (or plain majority) inducibility, and *unit margin*. The former is the standard notion of inducibility, and the latter asks that every arc be carried by exactly three voters against two. Only an unrestricted refutation bounds $N(5)$. However, the lemma below shows that for the Paley family and most tournaments derived from it, the unrestricted inducibility is equivalent to inducibility with margin at most 3 (i.e. support at most 4).

We also use unit margin as a fast filter for unrestricted inducibility. A unit-margin witness is in particular a majority witness, and the unit-margin search is much faster (Appendix F.4), so when the question is only whether a tournament is 5-inducible we search at unit margin first and at plain majority only if no unit-margin witness exists. The order-12 analysis of Appendix A works this way, and so do several of the family searches on 21 vertices (Appendix C). A tournament that fails the filter but has a majority witness is then a unit-margin obstruction found as a by-product, and several of those reported below arose this way.

Lemma 6.1 (3-cycle bound). *Let $u \to v \to w \to u$ be a directed triangle of T and let $\pi_1, \dots, \pi_k$ be any linear orders with k odd. Then*

$$c(u \to v) + c(v \to w) + c(w \to u) \le 2k.$$

Consequently, in any majority witness, no arc lying in a directed triangle has unanimous support k.

Proof. A linear order cannot rank u above v, v above w and w above u simultaneously, so each voter agrees with at most two of the three arcs; summing over the k voters gives the inequality. If $c(u \to v) = k$, the other two supports sum to at most k, contradicting the majority requirement that each be at least $(k+1)/2$. ■

For $k = 3$ the second conclusion is Proposition 1 of Milosz, Hamel and Pierrot (2018), with a transitivity argument; we need it for every odd k, and in the summed form above.

6.1 Paley(23) and Paley(31) are not 5-inducible

Theorem 6.2. *P_{23} is not 5-inducible. Consequently $N(5) \le 23$.*

The proof is a search that runs to completion. The base is $\{1, 2, 3, 6, 12\}$, which induces the regular tournament on 5 vertices and gives 8,031 base states. They are reduced to 2,591 by Lemma 5.1(b), by requiring one

```
ALGORITHM 1.  Placement search for k-inducibility of T

Input :  tournament T on V;  k voters;  margin (majority or unit);  base B
         Q: chosen base states of B    // Lem 5.1(d), (e): sorted, one per H-orbit
         R: anchoring representatives  // Lem 5.1(c), (b): vertices, pairs, or none
Output:  a witness profile, or a refutation

MAIN
  for each base state in Q  (a lex-canonical assignment of the base B to k orders):
      S <- B
      D(v | S) <- fresh enumeration, for every v not in S
      if any D(v | S) is empty:  continue     // this base state is dead
      if DFS(S) succeeds:        return its witness
  return "not k-inducible"                    // no base state extends

DFS(S)
  if S = V:  return the profile               // all vertices placed
  if every vertex of R is placed and no voter is anchored:
      return failure                          // Lem 5.1(b),(c); at the root if R is in B
  v* <- argmin  |D(v | S)|   over v not in S  // MRV, recomputed at EVERY node
        ties broken toward  max | |out(v) & S| - |in(v) & S| |
  for each slot tuple p in D(v* | S):
      insert v* at p in each of the k orders;  S' <- S u {v*}
      for each w not in S':
          D(w | S') <- REFINE( D(w | S), p, w, v* )
          if D(w | S') is empty:  undo;  continue with the next p
      if DFS(S') succeeds:  return its witness
      undo                                    // arena pointer reset
  return failure

REFINE(D, q, w, u)                            // q = the slot tuple just used for u
  D' <- {}
  for each p in D:
      lift p to p' voter by voter:
          p_i <  q_i  ->  p'_i = p_i        // w stays before u
          p_i >  q_i  ->  p'_i = p_i + 1    // w stays after u, shifted by the insertion
          p_i == q_i  ->  BOTH are legal, so the tuple splits in two
      keep those lifts whose support  c_w(u) = #{ i : w before u }
      meets the requirement of the single new arc between w and u
  return D'
```

voter to rank either $(1, 3)$ or $(1, 6)$ as the first pair. None of them produces a witness. This statement is also formally certified in Appendix B.1.

Thus none of the $23!^5$ five-voter profiles on 23 vertices has P_{23} as its majority tournament. It is the first explicit tournament of order below 43 shown not to be 5-inducible, and the first whose non-inducibility a general-purpose solver could not settle.

The same method settles a larger case: P_{31} is not 5-inducible, over 21,009 base states with a different base, reduced to 4,007 by Lemma 5.1(b). Appendix F.4 shows how the runtime scales in q, and Appendix F.4 tabulates the cost of every computation in this paper. Settling P_{31} shows that our search decides 31-vertex instances on a single laptop within a day, which the SAT and ILP solvers we tried did not.

Arc-criticality implies vertex-criticality: reversing an arc incident to v leaves $T - v$ unchanged, and the restriction of a witness to $V \setminus \{v\}$ is a witness for $T - v$. Each cell of Table 3 therefore records only the

strongest property established, at each of the two margins. "Not vertex-critical" also rules out arc-criticality.

Table 3: The strongest criticality property established for each Paley tournament, at each margin.

q	unit margin	unrestricted
19	**arc-critical**	5-inducible, so criticality does not apply
23	not vertex-critical	**arc-critical** (Section 6.3)
31	not vertex-critical	**not vertex-critical** (Section 6.3)
43	not vertex-critical	**not vertex-critical** (Section 6.3)

P_{19} is the only member critical at unit margin, and it is critical in the strongest sense: reversing any one arc makes it unit-margin inducible, which by the implication also gives vertex-criticality. At the other end, P_{31} and P_{43} fail to be vertex-critical at *both* margins. The interesting cell is $q = 23$: it is *vertex-critical unrestricted*, but *not* vertex-critical at unit margin. That is, $P_{23} - v$ is 5-inducible, but not with unit margin.

6.2 Paley(19) is not 5-inducible with unit margin

That P_{19} is 5-inducible is due to Bachmeier et al. (2019). Our search recovers it independently and contributes the negative half: it is *not* 5-inducible with unit margin. This is a complete refutation over all 2,200 base states of the base $\{1, 2, 3, 4, 6\}$, and it is the second statement we certify formally in Appendix B.1. These two facts supply the separation in Section 1: no arc of P_{19} can be unanimous, by the 3-cycle bound above, so it is 5-inducible with margin at most 3 and not with unit margin.

6.3 Paley(23) is arc-critical, unlike Paley(31) and Paley(43)

Theorem 6.3. *P_{23} is arc-critical, and hence vertex-critical.*

As $\mathrm{Aut}(P_{23})$ is transitive on arcs, it suffices to reverse the arc $(1, 2)$. The result, T^e, is rigid, so the search scans all 8,031 base states without anchoring, and the witness it finds is checked independently: its five orders are permutations of the 23 vertices whose majority tournament agrees with T^e on all 253 arcs. The two notions come apart among the four vertex-transitive unit-margin obstructions on 21 vertices: all four are vertex-critical, but two have every arc critical, one exactly half and one none (Appendix H).

Theorem 6.4. *Neither P_{31} nor P_{43} is vertex-critical for 5-inducibility.*

Proof. $\mathrm{Aut}(P_q - v)$ is the stabilizer of v in $\mathrm{Aut}(P_q)$: an automorphism of $P_q - v$ preserves its two out-degrees, $(q-1)/2$ on $N^+(v)$ and $(q-1)/2 - 1$ on $N^-(v)$, hence those two sets, and so extends to P_q by fixing v. With $v = 0$ the stabilizer is $\{x \mapsto ax : a \in \mathrm{QR}\}$, whose vertex orbits are QR and its complement, so vertex anchoring (Lemma 5.1(c)) applies with one representative from each. Every arc of $P_q - v$ lies in at least $(q-3)/4$ directed triangles, so by Lemma 6.1 margin at most 3 is no restriction, and a complete search at that margin over all 8,031 base states of a five-vertex base inducing the regular tournament on five vertices, $\{1, 2, 3, 4, 13\}$ in $P_{31} - v$ and $\{1, 2, 3, 4, 11\}$ in $P_{43} - v$, finds no witness. ■

This reconfirms Theorem 4.1 independently, since a witness for P_{43} would restrict to one for $P_{43} - v$.

6.4 Other structured tournament families

Beyond the Paley family we swept the structured families catalogued by McKay (2026) exhaustively for $n < 23$, and two families of order 23 completely; Table 4 gives the verdicts. A tournament is *locally transitive* if every in- and out-neighbourhood induces a transitive subtournament.

Table 4: Verdicts for five voters on the structured families; a dash marks a regime we did not settle. The arc reversals are one per arc orbit.

family	order	unrestricted	unit margin
vertex-transitive	$n < 23$	all inducible	all but P_{19} and 4 of 110 at $n = 21$
those 4, one arc reversed	21	all 40 inducible	—
doubly regular	19	both inducible	neither inducible
regular	≤ 15	all inducible	all inducible
self-converse	≤ 13	all inducible	all inducible
vertex-transitive, not P_{23}	23	all inducible	—
doubly regular	23	none of the 37	none of the 37
locally transitive	any	all inducible	all inducible, for every odd k

Since only an unrestricted refutation can move $N(5)$, the vertex-transitive family is closed as a source of candidates below 23 vertices. The other doubly regular tournament on 19 vertices is arc-critical at unit margin like P_{19}: its 171 arcs fall into 57 orbits of size 3, and reversing a representative of each yields a tournament with a unit-margin witness, 57 orbits and 57 witnesses. The searches over the $R_{15} = 18{,}400{,}989{,}629$ regular tournaments on 15 vertices and the $S_{13} = 95{,}458{,}560$ self-converse ones on 13 vertices cover these counts in full. Every vertex-transitive tournament on 23 vertices other than P_{23} is 5-inducible, while none of the 37 doubly regular ones is, at unit margin or unrestricted; up to converse they form 19 classes, each settled by a complete search (Appendix F.4). We do not search McKay's locally transitive tournaments: our companion paper shows that all are k-inducible with unit margin for every odd $k \geq 3$.

Taken together with the Paley results above and the families of Appendix C, these searches turn up 75 tournaments, up to isomorphism and converse, that five voters can produce but not with unit margin: 2 doubly regular ones at order 19; at order 21, 3 vertex-transitive ones (the four of Section 6.3, two of which are converses of each other), 15 arc reversals of two of them (Appendix H) and 42 near-doubly regular ones (Appendix C); at order 22, $P_{23} - v$; and at order 23, 11 circulant ones and P_{23} with one arc reversed.

6.5 Eight tournaments on 22 vertices are not 5-inducible

None of the 37 doubly regular tournaments on 23 vertices in the catalogue of McKay (2026) is 5-inducible (Table 4), which makes their one-vertex deletions natural candidates. We write Δ_i for the i-th, with vertices $v_1, \ldots, v_{23}$ in the catalogue's labelling, and $\Delta_i - v_j$ for Δ_i with v_j deleted.

Theorem 6.5. *The tournaments $\Delta_1 - v_1$, $\Delta_1 - v_2$, $\Delta_1 - v_{10}$, $\Delta_6 - v_5$, $\Delta_{11} - v_2$, $\Delta_{11} - v_5$, $\Delta_{11} - v_{20}$ and $\Delta_{28} - v_5$, on 22 vertices, are not 5-inducible. In particular, no tournament obtained from Δ_1 by deleting one vertex is 5-inducible. Consequently $N(5) \leq 22$.*

The second statement holds because $\mathrm{Aut}(\Delta_1)$, of order 55, has three vertex orbits, of sizes 11, 11 and 1, containing v_1, v_2 and v_{10}. The eight are pairwise non-isomorphic, even up to converse, and each has eleven vertices of out-degree 10 and eleven of out-degree 11; every arc lies in at least five directed triangles, so by the 3-cycle bound no arc of a witness is unanimous. Appendix G gives their adjacency matrices.

Deleting a vertex v of a doubly regular tournament on 23 vertices leaves two regular *halves* on 11 vertices, $N^+(v)$ and $N^-(v)$, each out-neighbour of v beating 6 in-neighbours and each in-neighbour 5 out-neighbours. We call v a *bisector* if its halves are isomorphic, or converse, to each other (an *isomorphic* or a *converse* bisector), and a tournament *bisectable* if it has a bisector or arises from one by deleting a bisector; one tournament on half the vertices then fixes both halves, which makes bisectable tournaments much easier to enumerate. Every doubly regular tournament on at most 23 vertices is bisectable, whereas 118 of the 722 on 27 vertices are not. Our obstructions cover every case. In $\Delta_1 - v_{10}$ and $\Delta_{11} - v_{20}$ both halves are P_{11}; in the four other deletions of Δ_1 and Δ_{11} they are isomorphic to each other and to each other's converse, with automorphism groups of order 5; in $\Delta_{28} - v_5$ they are converses of each other, with trivial automorphism groups; and in $\Delta_6 - v_5$ they are neither, and no other split works, so $\Delta_6 - v_5$ is non-bisectable.

Figure 1 shows $\Delta_1 - v_{10}$ as two copies of P_{11} on $X = \{x_0, \ldots, x_{10}\}$ and $Y = \{y_0, \ldots, y_{10}\}$ joined by a ladder. Its automorphism group, of order 55, has vertex orbits X and Y, and the automorphisms multiplying every index by a non-zero square fix x_0 and y_0 and form a subgroup H of order 5 mapping $B = \{x_1, x_3, x_4, x_5, x_9\}$, the out-neighbours of x_0 in X, onto itself. B induces the regular tournament on five vertices, whose 8,031 base states H groups into 1,611 orbits, so Lemma 5.1(e) keeps one per orbit while anchoring on x_0 and y_0 (Lemma 5.1(c)); the complete search over the 1,611 finds no witness. The certificate (Appendix B.1) uses the quotient without anchoring on $B' = \{y_1, y_3, y_4, y_5, y_9\}$, the in-neighbours of y_0 in Y, again keeping 1,611 of the 8,031. The other five deletions with an automorphism group of order 5, two of Δ_1 and the three of Δ_{11}, whose group fixes v_2, v_5 and v_{20}, are refuted in the same way over 1,611 chosen base states each, without certificates. $\Delta_6 - v_5$ and $\Delta_{28} - v_5$ have trivial automorphism groups, so their searches cover all 8,031 base states; $\Delta_6 - v_5$ is also certified (Appendix B.1).

All eight obstructions are *vertex-critical*, so none yields a 21-vertex obstruction by a further deletion, and the wider search below 22 vertices (Appendix C) has found none.

A ninth obstruction. The 231 arcs of $\Delta_1 - v_{10}$ fall into five orbits: inside X, inside Y, the rungs $y_i \to x_i$, the other arcs from Y to X, and the arcs from X to Y. Reversing a rung yields a 5-inducible tournament, but reversing an arc $y_j \to x_i$ with $j - i \in \mathrm{QR}$ does not: the result is rigid, and a complete search over all 8,031 base states finds no witness. This ninth obstruction is not isomorphic to any of the eight or to their converses, and it is vertex-critical: deleting an endpoint of the reversed arc leaves a deletion of $\Delta_1 - v_{10}$, and the other 20 deletions have explicit witnesses. Whether reversing an arc of the remaining three orbits restores 5-inducibility is open, as is the arc-criticality of the other seven; Section 7 collects the open questions.

7. Discussion and open problems

In this section we state the questions that our results leave open (Section 7.1) and describe how every result can be reproduced (Section 7.2).

7.1 Conjectures and open problems

Question 1. *Is $N(5) = 22$, that is, is every tournament on 21 vertices 5-inducible?*

The first version of this paper (arXiv:2609.13924v1), whose upper bound was $N(5) \le 23$, conjectured that $N(5) = 23$. The obstructions of Section 6.5 refute that conjecture, so we pose the question one order lower.

Both ends of $13 \le N(5) \le 22$ are constructive, unlike the counting bound of Section 3. The unit-margin obstructions at $n = 19$ and 21 cannot move the upper one, since every one of them is majority-inducible. The evidence below 22 comes mostly from the descendants of the doubly regular tournaments on 23 vertices (Appendix C): every tournament on 20 vertices obtained from one of them by deleting three vertices is 5-inducible, and so is every one on 21 vertices below any of the nine 22-vertex obstructions.

Every class we searched below 22 vertices was chosen because symmetry, a construction, or descent from a known obstruction made it searchable, and Appendix C lists them all. These classes cluster around the doubly regular tournaments, where every obstruction we know of lies. What they show is that no obstruction on 21 vertices lies *among them*; they say little about the roughly 3×10^{43} tournaments on 21 vertices.

Conjecture 2. *P_q is not 5-inducible for any eligible prime $q \ge 23$, i.e. $q \equiv 3 \pmod 4$.*

It holds at $q = 23$, 31 and 43, and only finitely many q are in doubt: $\alpha^*(P_q) = \mathrm{MAS}(P_q)/C$ tends to $1/2$, because Paley tournaments are quasi-random in the sense of Chung and Graham (1991, Section 6), and one of their equivalent quasi-random properties (P_9) is that every linear order of the vertices agrees with $(1/2 + o(1))\binom{q}{2}$ arcs. Hence the predictability eventually falls below $3/5$, past which the necessary condition $\alpha^* \ge 3/5$ refutes 5-inducibility with no search at all. What is missing is a bound on where that range ends; the values in Table 2 fall too irregularly to extrapolate one.

Open problem 1 (vertex- and arc-criticality). Whether P_q minus *two* vertices is 5-inducible, equivalently whether P_q minus one vertex is vertex-critical, is open at $q = 31$ and $q = 43$. For $q = 43$, $P_{43} - 2v$ is unique up to isomorphism, since $\mathrm{Aut}(P_{43})$ acts transitively on unordered pairs of vertices, and the screen of Section

4.2 extends to it. The two deleted vertices can be chosen after the screen rather than before: a pair of orders survives the deletion exactly when some set of at most two vertices meets every cyclic triangle that both orders double-back, a property invariant under Aut, so the reduction by a factor of 903 survives the deletion. The condition on the two voters that agree with the most arcs sharpens to $\delta_1 + 4\delta_2 \leq 5\,\mathrm{MAS} - 3C$, where δ_1, δ_2 are their levels; the right-hand side is 45 for $P_{43} - 2v$.

Open problem 2 (the margin hierarchy). The first of unit margin $\subseteq$ margin $\leq 3 \subseteq$ unrestricted is strict (Sections 6.2 and 6.4); whether the second ever is, that is, whether some tournament is inducible by five voters but not at margin ≤ 3, is open. Lemma 6.1 forces such a tournament to have an arc in *no* directed triangle, which rules out every regular tournament. Conjecture 8.1 of our companion paper states that the inclusion is strict at every odd $k > 3$.

Conjecture 3. *For every odd k, every k-inducible tournament is $(k+2)$-inducible with unit margin.*

Conjecture 3 extends Conjecture 8.1 of our companion paper, which concerns the classes for a single k, to classes across different k. Adding an order together with its reverse raises every support by one, so k-inducibility with margin at most M implies $(k+2)$-inducibility with margin at most M. Conjecture 3 strengthens this to unit margin, and with the inclusions within each k it would nest every class at a lower k inside every class at a higher k, whatever the margins. It trivially holds at $k = 1$, since a transitive tournament is the majority of an order taken twice and its reverse once, with every arc carried two against one. At $k = 3$, Appendix A shows that every tournament on at most 12 vertices is 5-inducible with unit margin, so a counterexample needs at least 13 vertices. At $k = 5$, both doubly regular tournaments on 19 vertices, which are 5-inducible but not with unit margin, are 7-inducible with unit margin.

7.2 Reproducibility

The code, data and verdict files are at https://github.com/Leonardini/TournamentsBeyond5Voters. Its `CLAIMS.md` carries one row per statement made here, naming the artifact that holds the verdict and the command that re-derives it, and it ends with an explicit list of what the package does not establish. A permanent archive of the accepted version will be deposited on acceptance, with its DOI recorded here.

The repository contains the search engine as a single C translation unit, the SAT encoding and cube tooling in Python, the certification driver, the independent witness verifiers, the tournaments as bit strings, and a verdict record for every result reported here. Each verdict records the exact command line, the base and break used, the coverage audit (the check that the cubes miss no base state, Appendix B.2), the runtime, and the two root hashes for the formally certified results. Three further folders hold the counting bounds (`counting_bounds/`), the proof for P_{43}, which rebuilds from one script (`paley43/`), and every check that the tournaments on at most 12 vertices are 5-inducible with unit margin (`n12_unit_margin/`). The last ships explicit profiles for the non-3-inducible tournaments on 8 and 9 vertices; at larger orders the same code regenerates them, for a single tournament in milliseconds and for all tournaments of an order in minutes at $n = 10$, in about 44 core-hours at $n = 11$ and in 2,616 cluster core-hours at $n = 12$.

Of the two root hashes published with each certified computation, ROOT (CNF) commits to the SHA-256 of every cube's CNF and must be reproduced exactly by anyone who regenerates the cube set, whereas ROOT (proofs) also commits to the LRAT proof bytes, which depend on the solver build, so a mismatch there is not evidence of an error. A rerun of both refutations on a compute cluster, under a different compiler and CaDiCaL build, matched both CNF roots, and both coverage instances rebuilt to their committed CNF hashes and were refuted again. Search verdicts replicate under any correct implementation, node counts replicate exactly when the tie-breaking rule, the vertex numbering, the base and the margin are held fixed (the engine prints all but the numbering on every run, and the repository stores the tournaments themselves), and core-hours replicate only in order of magnitude (Appendix F.4).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

LC acknowledges funding from the MRC Centre for Global Infectious Disease Analysis (reference MR/X020258/1), funded by the UK Medical Research Council (MRC). This UK funded award is carried out in the frame of the Global Health EDCTP3 Joint Undertaking.

CRediT authorship contribution statement

Leonid Chindelevitch: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review and editing. **Ararat Harutyunyan:** Conceptualization, Formal analysis, Writing – review and editing.

Data availability

The code, data and verdict files supporting the findings of this study are at https://github.com/Leonardini/TournamentsBeyond5Voters, described in Section 7.2.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) for exploratory reasoning, implementation assistance, and drafting and editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Acknowledgements

This work was granted access to the HPC resources of IDRIS under the allocation 2025-AD010616623R1 made by the Grand Équipement National de Calcul Intensif (GENCI).

References

Bachmeier G, Brandt F, Geist C, Harrenstein P, Kardel K, Peters D, Seedig HG (2019) k-Majority digraphs and the hardness of voting with a constant number of voters. *Journal of Computer and System Sciences* 105:130–157

Biere A, Faller T, Fazekas K, Fleury M, Froleyks N, Pollitt F (2024) CaDiCaL 2.0. In: *Computer Aided Verification (CAV 2024)*, LNCS 14681, Springer, pp 133–152. doi:10.1007/978-3-031-65627-9_7

Brakensiek J, Heule M, Mackey J, Narváez D (2020) The resolution of Keller's conjecture. In: *Automated Reasoning (IJCAR 2020)*, LNCS 12166, Springer, pp 48–65

Chindelevitch L, Harutyunyan A (2026) Tournaments not inducible by three voters. Submitted to *Mathematical Social Sciences* jointly with this paper

Chung FRK, Graham RL (1991) Quasi-random tournaments. *Journal of Graph Theory* 15(2):173–198

Cruz-Filipe L, Heule MJH, Hunt WA Jr, Kaufmann M, Schneider-Kamp P (2017) Efficient certified RAT verification. In: *Automated Deduction (CADE 2017)*, LNCS 10395, Springer, pp 220–236

Cruz-Filipe L, Marques-Silva J, Schneider-Kamp P (2019) Formally verifying the solution to the Boolean Pythagorean triples problem. *Journal of Automated Reasoning* 63(3):695–722. doi:10.1007/s10817-018-9490-4

Dummit DS, Foote RM (2004) *Abstract Algebra*, 3rd edn. Wiley, Hoboken, NJ

Dushnik B, Miller EW (1941) Partially ordered sets. *American Journal of Mathematics* 63(3):600–610

Eggermont C, Hurkens C, Woeginger GJ (2013) Realizing small tournaments through few permutations. *Acta Cybernetica* 21(2):267–271

Erdős P, Moser L (1964) On the representation of directed graphs as unions of orderings. *Publications of the Mathematical Institute of the Hungarian Academy of Sciences* 9(1–2):125–132

Fidler D (2011) A recurrence for bounds on dominating sets in k-majority tournaments. *The Electronic Journal of Combinatorics* 18(1):#P166. doi:10.37236/653

Fiol MA (1992) A note on the voting problem. *Stochastica* 13(1):155–158

Held M, Karp RM (1962) A dynamic programming approach to sequencing problems. *Journal of the Society for Industrial and Applied Mathematics* 10(1):196–210. doi:10.1137/0110015

Heule MJH, Kullmann O, Marek VW (2016) Solving and verifying the Boolean Pythagorean triples problem via cube-and-conquer. In: *Theory and Applications of Satisfiability Testing (SAT 2016)*, LNCS 9710, Springer, pp 228–245

Heule MJH, Kullmann O, Wieringa S, Biere A (2012) Cube and conquer: guiding CDCL SAT solvers by lookaheads. In: *Haifa Verification Conference (HVC 2011)*, LNCS 7261, Springer, pp 50–65

Heule MJH (2018) Schur number five. In: *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2018)*, pp 6598–6606

Kahn AB (1962) Topological sorting of large networks. *Communications of the ACM* 5(11):558–562. doi:10.1145/368996.369025

McGarvey DC (1953) A theorem on the construction of voting paradoxes. *Econometrica* 21(4):608–610

McKay BD (2026) Combinatorial data: digraphs (tournament catalogues). Accessed 10 August 2026. https://users.cecs.anu.edu.au/~bdm/data/digraphs.html

McKay BD, Piperno A (2014) Practical graph isomorphism, II. *Journal of Symbolic Computation* 60:94–112

Milosz R, Hamel S, Pierrot A (2018) Median of 3 permutations, 3-cycles and 3-hitting set problem. In: *Proceedings of IWOCA 2018*, LNCS 10979, Springer, pp 224–236. doi:10.1007/978-3-319-94667-2_19

OEIS Foundation Inc. (2026a) Entry A000568, Number of outcomes of unlabeled n-team round-robin tournaments. The On-Line Encyclopedia of Integer Sequences. https://oeis.org/A000568

OEIS Foundation Inc. (2026b) Entry A002785, Number of self-complementary oriented graphs with n nodes. The On-Line Encyclopedia of Integer Sequences. https://oeis.org/A002785

OEIS Foundation Inc. (2026c) Entry A096368, Number of unlabeled regular tournaments with $2n+1$ nodes. The On-Line Encyclopedia of Integer Sequences. https://oeis.org/A096368

Pollitt F, Fleury M, Biere A (2023) Faster LRAT checking than solving with CaDiCaL. In: *Theory and Applications of Satisfiability Testing (SAT 2023)*, LIPIcs 271, pp 21:1–21:12. doi:10.4230/LIPIcs.SAT.2023.21

Schrijver A (1983) Bounds on the number of Eulerian orientations. *Combinatorica* 3(3–4):375–380. doi:10.1007/BF02579193

Shepardson D, Tovey CA (2009) Smallest tournaments not realizable by 2/3-majority voting. *Social Choice and Welfare* 33(3):495–503. doi:10.1007/s00355-009-0375-7

Appendix A. The order-12 case analysis

Every tournament on 12 vertices is 5-inducible, so $N(5) \geq 13$. This appendix describes that computation, which starts from a census of order 11. It is one of the results in this paper that was not run on the laptop of Section 6: it ran on a shared cluster, and its runtime is converted as Appendix F.4 describes.

A.1 Every tournament on eleven vertices is 5-inducible

Our first census shows that every tournament on at most 11 vertices is 5-inducible, so $N(5) \geq 12$. Through $n = 10$ each tournament is decided by the integer program of Appendix F.1. At $n = 11$ no integer program was needed: of the $D_{11} = 903{,}753{,}248$ tournaments, 382,625,248 are 3-inducible by the triple-local test of our

companion paper, and every one of the remaining 521,128,000 admits a *unit-margin* five-voter profile, found by a constraint search and re-checked independently. A tournament with a non-3-inducible subtournament is itself non-3-inducible, and restricting a profile to a subset of the vertices preserves every pairwise split, so every non-3-inducible tournament on at most 11 vertices is 5-inducible with unit margin.

A.2 Reduction to one-vertex extensions

A direct census is out of reach: $D_{12} \approx 1.5 \times 10^{11}$ classes, 170 times the order-11 census of Appendix A.1. We use that census as the outer loop. We write $L + S$ for the tournament obtained from an order-11 tournament L by adding a vertex v whose out-neighbourhood is $S \subseteq V(L)$. Two observations simplify the analysis.

(i) Every order-12 tournament is $L + S$ for some order-11 L and some S.

(ii) Restricting a profile that realizes $L + S$ to $V(L)$ gives a profile that realizes L.

By (i), enumerating all D_{11} order-11 classes and settling all 2,048 extensions of each settles every order-12 tournament; all of the former are 5-inducible by the census of Appendix A.1, which is what makes the outer loop available. By (ii), a profile realizing $L + S$ may be sought by *inserting* v into a profile realizing L with no loss of generality, since every profile for $L + S$ arises that way. That is what phase 1 uses.

A.3 Three phases

The census runs in three phases, fastest first.

Phase 1, the covering search. Rather than 2,048 separate instances per order-11 tournament, one search enumerates profiles realizing L *at unit margin* and, for each, records every out-neighbourhood obtainable by inserting v into its five orders: v's position in each order is a threshold, and the support of $v \to u$ is the number of orders that place v above u, so one traversal certifies many extensions at once. This is the engine of Section 5.2 with v's placement read off rather than fixed in advance. The margin makes this phase pay twice: it keeps the enumeration of profiles for L small, and a unit-margin witness is in particular a majority witness, so every extension the phase covers is settled in both regimes at once. The phase is one-sided: an extension it covers is certified 5-inducible, but one left over is not refuted. This phase settles every one of the 2,048 extensions of 99.5% of the order-11 tournaments.

Phase 2, deciding what is left. Each extension phase 1 leaves open is materialized as an order-12 tournament $L + S$ and decided by the same engine, again at unit margin, where a witness settles it positively. About one extension in 200,000 reached this phase, and every one of them is 5-inducible with unit margin.

Phase 3, plain majority. Only a *majority* refutation would be a counterexample, so any instance negative at unit margin is re-run without the margin restriction. Phase 3 was never invoked: every tournament was settled at unit margin, so the weaker question never had to be asked.

A.4 One tournament of each converse pair

T is 5-inducible if and only if its converse $\bar{T}$ is (Section 1). Moreover $\overline{L + S} = \bar{L} + (V(L) \setminus S)$, so settling all 2,048 extensions of L settles all 2,048 of $\bar{L}$ as well. We therefore keep one tournament of each converse pair, by the rule that L is kept if and only if $\text{canon}(L) \leq \text{canon}(\bar{L})$. This keeps exactly one of each pair together with every self-converse tournament, $(D_{11} + S_{11})/2$ in all, with $S_{11} = 279{,}968$; this saves a factor of $\lessapprox 2$. We also check the number of tournaments kept is $(D_{11} + S_{11})/2$.

Total runtime: 2,615.6 core-hours. The search code, the per-part records and the roll-up are at `n12_unit_margin/n12_cover` in the repository of Section 7.2.

Appendix B. The formal certificates

In this appendix we describe how the four certified negative results are proved (Appendix B.1) and what the certificates do and do not establish (Appendix B.2).

B.1 The certified refutations

The four certified refutations follow an established design, and we claim no methodological novelty. *Cube-and-conquer* (Heule, Kullmann, Wieringa and Biere 2012) splits a hard instance into partial assignments, the *cubes*, each handed to a CDCL solver; the standard of proof is that of the Boolean Pythagorean triples problem (Heule et al. 2016), later re-checked with a formally verified checker (Cruz-Filipe et al. 2019), of Schur number five (Heule 2018) and of the Keller conjecture (Brakensiek et al. 2020), in which every cube's refutation is validated by an independent checker. Each cube is refuted by CaDiCaL 2.0.0 (`--lrat=true --checkproof=0`), which writes an LRAT proof (Cruz-Filipe et al. 2017): each derived clause lists the earlier clauses it follows from, so `lrat-trim` (Pollitt et al. 2023) checks it without search. Our cubes are base states, a combinatorial decomposition, so the coverage of the cube set is certified separately (Appendix B.2). Where Heule et al. (2016) published a cube certificate from which the refutations can be regenerated, at about 13,000 core-hours to regenerate and 16,000 to check, we *verify and discard*: each LRAT proof is checked, hashed and deleted, and only the hash is published (Section 7.2). Table B.1 summarizes the four certificates.

Table B.1: The four certified refutations. The cubes solved are those left by the arc-orbit breaks (P_{19}, P_{23}) or the quotient ($\Delta_1 - v_{10}$); bases are in each tournament's labelling; core-hours cover solving and checking; the last column is the time to solve the coverage instance.

refutation	base	cubes solved	core-hours	coverage
P_{19}, unit margin	$\{1, 2, 3, 4, 5, 12\}$	22,876 of 142,251	25.1	7.56 s
P_{23}	$\{1, 2, 3, 4, 6, 7\}$	343,896 of 3,414,729	236.0	20 min
$\Delta_6 - v_5$	$\{1, 2, 3, 4, 13, 19\}$	all 742,374	1,292.7	82.8 s
$\Delta_1 - v_{10}$	$\{13, 15, 16, 20, 22\}$	1,611 of 8,031	≈ 732	0.08 s

For P_{19} at unit margin the dissent-Boolean encoding of Appendix F.1 has 855 variables and 12,255 clauses; for $\Delta_6 - v_5$ and $\Delta_1 - v_{10}$ it is taken at $n = 22$ and plain majority: 1,155 variables and 17,710 clauses. The coverage half is a single unsatisfiable instance that negates every base state, since coverage must see all of them whereas the orbit breaks apply only to the search half. For P_{23} it has 350 constraint clauses, 356 symmetry-breaking clauses and 3,414,729 negated cubes, and its LRAT proof takes 2.05 GiB; for $\Delta_6 - v_5$ it is the same construction with 742,374 negated cubes. $\Delta_6 - v_5$ is the most expensive certificate although it is smaller than P_{23}: with no automorphisms every base state reaches the solver, against one in ten for P_{23}.

B.2 What the certificates do and do not establish

The trust chain has five links:

Table B.2: The five links of the trust chain.

link	status
every live cube is unsatisfiable	machine-checked: CaDiCaL, then `lrat-trim`
the cubes cover every lex-canonical base assignment	machine-checked: LRAT, verified by `lrat-trim`
(L1) orbit anchoring loses no generality	human-checked: Lemma 5.1(b)
(L2) lex-ordering the voters loses no generality	human-checked: Lemma 5.1(d)
(L3) one base state per orbit loses no generality	human-checked: Lemma 5.1(e)

Appendix F.2 proves the human-checked links, which nothing in the pipeline checks. P_{19} and P_{23} use (L1) and (L2), $\Delta_1 - v_{10}$ uses (L2) and (L3), and $\Delta_6 - v_5$ uses (L2) alone. Table B.3 says what decided each result and what independent check, if any, stands behind it. A negative verdict from the depth-first engine alone rests on the engine's correctness; a positive one never does, as a separate program checks each witness.

Table B.3: What each result rests on. The first six rows are refutations (not 5-inducible, at unit margin where stated); "engine" is the depth-first search of Appendix F, run twice on P_{23} with different bases and anchors, and a verifier, sharing no code with the search, checks a profile's majority tournament against the target.

result	decided by	independent check
P_{23}	engine, twice	LRAT certificate
$\Delta_1 - v_{10}$, $\Delta_6 - v_5$	engine	LRAT certificate
P_{19}, at unit margin	engine	LRAT certificate
the other 22-vertex obstructions	engine	none
P_{27}, P_{31}, $P_{31} - v$	engine	none
P_{43}	proof, two programs	$P_{43} - v$, by the engine
all orders ≤ 11 inducible	ILP, constraint search	verifier, per profile
all of order 12 inducible	extension search	check inside the search
positive results of Section 6	engine	verifier, per witness
counting bounds	proof	none needed

Appendix C. The search below 22 vertices

This appendix summarizes the search for an obstruction on fewer than 22 vertices, which would improve the upper bound of Section 6.5. It has found none.

The search runs downward, because 5-inducibility is inherited by subtournaments: restricting a witness for T to a subset of the vertices gives a witness for the subtournament. An obstruction on fewer vertices therefore lies below some larger one, and the natural place to look is among the vertex deletions of known obstructions. A tournament is settled positively as soon as one base state yields a witness, which is cheap, and negatively only by a complete search, which is not; so the tournaments of each order are first screened for witnesses, and only tournaments that survive the screen are searched further.

The starting points are the 36 doubly regular tournaments on 23 vertices other than P_{23}, all of which are obstructions (Section 6.4). P_{23} itself is excluded, since $P_{23} - v$ is 5-inducible (Section 6.3) and so, by inheritance, is every tournament below it. Table C.1 lists them up to isomorphism and converse.

Table C.1: Tournaments below the 36 doubly regular ones on 23 vertices, up to isomorphism and converse.

vertices	deleted	classes	status
22	1	378	8 searched completely, all obstructions (Section 6.5)
21	2	4,108	1,192 settled, all 5-inducible
20	3	28,724	all 5-inducible

At 22 vertices all 378 classes were first screened at up to 50 base states each, without a single witness, and the eight complete searches were chosen from among them. At 21 vertices the 1,192 settled classes include all 506 two-vertex deletions of Δ_6 and Δ_{28}, so every 21-vertex subtournament of either parent is 5-inducible, and in particular $\Delta_6 - v_5$ and $\Delta_{28} - v_5$ are vertex-critical. The other settled classes are a stratified random sample of 400; all 53 classes below Δ_{11}; every class obtained by deleting v_5 together with one other vertex, the deletion that produced those two obstructions; and 19 classes settled by inserting the missing vertex into a known witness of one of their 20-vertex subtournaments, as in phase 1 of Appendix A.3; the remaining 2,916 classes are open. At 20 vertices every class is 5-inducible, and so, by inheritance, is every tournament on 19 vertices or fewer below the 36.

Two obstructions on 25 vertices from other families were descended the same way: P_{27} with two vertices deleted, and the circulant tournament on 25 vertices in which every arc lies in six directed triangles. Neither is 5-inducible, by complete searches, and every 21-vertex subtournament of either is 5-inducible, so neither family yields an obstruction below 22.

Other families on 21 vertices. We also searched families on 21 vertices that a symmetry or a construction makes enumerable, all of them close to doubly regular. A regular tournament on 21 vertices cannot be doubly regular, since $(21-3)/4$ is not an integer; we call it *near-doubly regular* if every pair of vertices has 4 or 5 common out-neighbours. The table below lists every family we searched on 21 vertices, again up to isomorphism and converse, with the number of classes settled; the vertex-transitive tournaments on 21 vertices, a family enumerated in full, are covered in Section 6.4. Every settled class is 5-inducible, by a witness verified independently of the search; none was settled negatively.

Table C.2: The families searched on 21 vertices, up to isomorphism and converse.

family on 21 vertices	classes	settled
two vertices deleted from a doubly regular tournament on 23 vertices	4,108	1,192
four vertices deleted from the circulant tournament on 25 vertices	286	286
six vertices deleted from P_{27}	150	150
near-doubly regular, rigid, drawn at random	78	78
near-doubly regular, with a bisector and an automorphism of order 5	28	28
near-doubly regular, with a fixed-point-free automorphism of order 3	31	31

The 31 classes of the last family are likely, but not confirmed, to be all of it: the enumeration found no new class after its first 225 solutions, and we stopped it at 4,567 without running it to completion. Two of the 31 are bisectable and the other 29 are not. In the two families with an automorphism, several classes have only majority witnesses: 12 of the 28 with a bisector, and 30 of the 31 with an automorphism of order 3.

Appendix D. The MAS algorithm: orbit subset-DP, meet-in-the-middle, and the negation/converse join key

This appendix describes the dynamic program that certifies $\mathrm{MAS}(P_{43}) = 543$ (Section 4.1) and enumerates the level-≤ 1 shell (Section 4.2).

D.1 Base recurrence (Held–Karp 1962). For $S \subseteq V$ let $g(S) = \mathrm{MAS}(T[S])$. The last vertex v of an order of S gains its in-arcs from $S \setminus \{v\}$, so

$$g(\varnothing) = 0, \qquad g(S) = \max_{v \in S} \big[\, g(S \setminus \{v\}) + |\{u \in S \setminus \{v\} : u \to v\}| \,\big],$$

and $\mathrm{MAS}(T) = g(V)$.

D.2 Automorphism quotient. As $T[\sigma S] \cong T[S]$ for $\sigma \in \mathrm{Aut} = \{x \mapsto ax + b : a \in \mathrm{QR}\}$, we keep one representative per orbit: 4.9×10^9 states up to $|S| = 21$ (almost 903 times fewer). The representative is the minimum bit-mask over the images that send a vertex of maximum out-degree within S to 0. Layer $j+1$ comes from layer j: a stored S yields, for each $v \notin S$, the record $(\mathrm{canon}(S \cup \{v\}),\ g(S) + |\{u \in S : u \to v\}|)$, with the gain computed before canonicalizing; equal keys keep the largest value, $g(S \cup \{v\})$.

D.3 Meet-in-the-middle at the equator. Storing every layer up to $|S| = 43$ is out of reach, so we build layers 0 to 22 only and split every order in the middle. Let π be an order of V and S the set of its first 21 vertices (the **equator** is $h = 21$). An arc of T lies inside S, inside $V \setminus S$, or between the two, and an arc between them is forward in π exactly when it goes from S to $V \setminus S$. Hence

$$\mathrm{fwd}(\pi) = \mathrm{fwd}(\pi|_S) + \mathrm{cross}(S) + \mathrm{fwd}(\pi|_{V \setminus S}),$$

where $\mathrm{cross}(S)$ counts the arcs from S to $V \setminus S$, and maximizing the two halves separately gives

$$\mathrm{MAS}(T) = \max_{|S|=21} \big[\, g(S) + \mathrm{cross}(S) + g(V \setminus S) \,\big].$$

The first term is read from layer 21. The suffix $V \setminus S$ has 22 vertices, but the tables were built from the front of an order, so we read the suffix through a twin order. Negation $x \mapsto -x$ maps T onto its converse, because

-1 is a non-residue; so reversing π and negating every vertex gives an order π^* with $\mathrm{fwd}(\pi^*) = \mathrm{fwd}(\pi)$, whose first 22 vertices form the set $-(V \setminus S)$. In particular $g(V \setminus S) = g(-(V \setminus S))$, which layer 22 stores under the key $\mathrm{canon}(-(V \setminus S))$.

D.4 The join: the negation/complement commutes with the quotient by Aut**.** Layer 21 holds one representative per orbit, so the maximum above must be taken over orbits, and the key of the suffix must not depend on which representative was stored. Let $\nu(S) = -S$, $\kappa(S) = V \setminus S$ and $\varphi = \nu \circ \kappa$, so that the suffix of S is looked up at $\mathrm{canon}(\varphi(S))$. (i) κ commutes with every bijection of V; (ii) $\nu \circ \sigma_{a,b} = \sigma_{a,-b} \circ \nu$, so negation normalizes Aut while lying outside it; hence (iii) $\varphi \circ \sigma_{a,b} = \sigma_{a,-b} \circ \varphi$. So if S and S' lie in one orbit, so do $\varphi(S)$ and $\varphi(S')$: $\mathrm{canon}(\varphi(S))$ is constant on orbits, and φ induces a one-to-one map from the orbits of 21-sets to the orbits of 22-sets. As g and cross are constant on orbits as well, the maximum over all 21-sets equals the maximum over the stored representatives.

We stream layer 21 and form, for each representative S, the pair $(\mathrm{canon}(\varphi(S)),\ g(S) + \mathrm{cross}(S))$; we sort these pairs by key and merge them with the stored layer-22 table, which is sorted by the same key. Each match gives $\mu(S) = g(S) + \mathrm{cross}(S) + g(\varphi(S))$, the largest number of forward arcs of an order whose first 21 vertices form S, and $\mathrm{MAS}(T) = \max_S \mu(S) = 543$. Backtracking through the stored values from the splits with $\mu(S) \geq 542$ lists every order of level ≤ 1, the shell of Section 4.2 (orbit counts in Appendix E).

Appendix E. Counts for the Paley(43) proof

This appendix gives the counts behind the proof that $N(5) \leq 43$ (Section 4): the structural counts of Paley(43) and the certified value MAS = 543 (Table E.1), then the sizes of the shell and of the screen.

Table E.1: Structural counts and certified quantities for Paley(43).

quantity	formula	value
arcs C	$q(q-1)/2$	903
cyclic triangles N_Δ	$(q^3-q)/24$	3311
$\lvert\mathrm{Aut}(P)\rvert$	$q(q-1)/2$	903
out-degree	$(q-1)/2$	21
triangles per arc	$(q+1)/4$	11
$\mathrm{MAS}(P)$	dynamic program (Appendix D), certified	543
$\alpha^* = \mathrm{MAS}/C$		181/301
slack $5\,\mathrm{MAS} - 3C$		6
forced top-two level	$\lfloor \mathrm{slack}/4 \rfloor$	1

The level-≤ 1 shell has 1,841,303 Aut-orbits and 1,662,696,609 orders: 19,651 orbits (17,744,853 orders) at level 0 and 1,821,652 orbits (1,644,951,756 orders) at level 1. In the screen (Section 4.2), the shell orders have 4,709,640 distinct restrictions to the cyclic triangles inside the window $\mathcal{W}$, and 5,092,111 pairs of these restrictions are disjoint. Only the 678,686 restrictions that belong to such a pair, and the 347,694,990 shell orders that have one of them, need the exact check. It compared 4,376,325,129 pairs of an orbit representative and such an order, and found no pair with disjoint double-back sets (0 of 4,376,325,129).

Appendix F. The search engine

Integer programming and satisfiability are both central to this project. Every optimum reported in the companion paper is certified by an integer or linear program, and the integer program of Appendix F.1 below is that paper's k-inducibility formulation with antisymmetry substituted out; the only machine-checked theorem in this paper is a SAT computation from end to end, with CaDiCaL (Biere et al. 2024; Pollitt et al. 2023) emitting LRAT proofs that `lrat-trim` (Pollitt et al. 2023) checks. The claim here is narrow: on this one decision problem, at these sizes, neither tool reaches the answer while a search built around the problem's own structure does, and the same SAT machinery is then indispensable for turning that answer into a proof.

F.1 Integer programming and SAT are both too slow

The natural integer program has one binary $x_{v,ij}$ per voter v and pair $i < j$, meaning that v ranks i above j, with $x_{v,ji}$ read as $1 - x_{v,ij}$; transitivity rows $0 \le x_{v,ij} + x_{v,jl} - x_{v,il} \le 1$ for $i < j < l$; and one majority row $\sum_v x_{v,ij} \ge (k+1)/2$ per arc. At $q = 23$, $k = 5$ it has 1,265 binaries, 17,710 transitivity rows and 253 majority rows. Setting $x_{v,ij} = 3/5$ on every arc of T and $2/5$ on every non-arc satisfies every row, so the linear programming (LP) relaxation is feasible and, the problem having no objective, gives branch-and-bound nothing to prune by; that point is invariant under both symmetry groups, so it gives no branching direction either. CPLEX and Gurobi both failed to close instances at $n = 19$ within a day.

The satisfiability encoding is *dissent-Boolean*: one variable per voter v and arc of T, true when v dissents from the arc. Each arc has at most $(k-1)/2$ dissents for majority and exactly $(k-1)/2$ for unit margin, and for every voter and triple the two dissent patterns that would make the triple cyclic are forbidden. For $k = 5$ that is $5\binom{n}{2}$ variables, 15 clauses per arc at unit margin and 10 per triple, and the certified refutations of Appendix B are built on it. On uniform random tournaments the `Sat-Check-k-Majority` of Bachmeier et al. (2019, Table 4) averages 0.23 seconds at $n = 18$ and 18.95 at $n = 23$, roughly doubling per vertex, whereas on P_{23} the same solver did not terminate within six weeks: random tournaments have no automorphisms to rediscover and many witnesses. Better encodings and explicit symmetry breaks help but do not suffice; supplying the decomposition into base states as cubes does (Section 5.1).

F.2 The symmetry reductions in full

In this subsection we prove Lemma 5.1 and explain how its parts are used.

Proof of Lemma 5.1. (a) Let w be a witness and choose $g \in G$ with $g\,\varphi(w) \in R$. Then gw is a witness, since permuting voters changes no margin and relabelling by an automorphism carries the margin of each arc to its image, and $\varphi(gw) = g\,\varphi(w) \in R$. Parts (b) and (c) are (a) with $G = \mathrm{Aut}(T)$, $\varphi(\pi)$ the top pair, respectively the top vertex, of π_1, and R the representatives. Part (d) is (a) with $G = S_k$, φ the identity and R the sorted profiles, every S_k-orbit containing one. A second application of (a) may use any group element that preserves what the first guaranteed, and permuting voters preserves "some voter ranks a_i first and b_i second" (or "v_i first"), which depends on the multiset of orders and not on which voter carries it; so (d) applies after (b) or (c). For (e), relabelling by $h \in H$ maps a base state of C to one, since $h(C) = C$ and h restricts to an automorphism of $T[C]$, while S_k acts trivially on base states, which are multisets; and the base state of C that $h\pi$ extends is h applied to the one π extends. So (a) with $G = S_k \times H$ and $\varphi(\pi)$ that base state gives a witness extending a chosen base state of C, and its orders restricted to B form a chosen base state of B that it extends. If H fixes the representatives, each $h \in H$ maps an anchored voter to an anchored voter, so the same argument applies after (b) or (c). ■

Part (b) guarantees only that *some* i works, not any particular one, and it is this weak form that lets (d) follow it; had it said "voter 1 ranks a_i first", the composition would fail. For a Paley tournament there are two orbits on ordered pairs, the arcs and the non-arcs: the maps $x \mapsto a(x - t)$ with $a \in \mathrm{QR}$ are automorphisms, they send (t, s) to $(0, a(s - t))$, and as a ranges over QR they leave exactly two orbits, separated by whether $s - t \in \mathrm{QR}$. On P_7 (Figure F.2) the representatives are $(1, 2)$ and $(1, 4)$. Otherwise we compute $\mathrm{Aut}(T)$ and its orbits with `nauty` 2.8.6 (McKay and Piperno 2014), which also provides every automorphism group, orbit set and canonical form reported in this paper; (b) and (c) are alternatives, and the better one depends on the numbers of orbits on pairs and on vertices.

When every representative pair lies inside B, a voter ranking a_i first and b_i second overall does so within B, which the base state reveals, so a base state in which no voter tops B with a chosen pair extends to no anchored witness and may be discarded; the survivors are the *live* base states. Otherwise the search tests the anchor once every representative has both endpoints placed: an insertion can destroy an anchor but never create one, so a partial profile with no anchored voter has no anchored extension. The SAT certification tests it only once, to decide which cubes reach the solver.

In part (e) the condition on the representatives cannot be dropped: if H moves one, mapping an anchored witness into a chosen base state can destroy its anchor. Automorphism groups of tournaments have odd order, since an automorphism of order 2 would swap the endpoints of some arc, and an automorphism h maps B

onto itself exactly when B is a union of its cycles; a 5-cycle of an automorphism of order 5 is such a base. When no subgroup maps a base of the size we use onto itself, $C \subsetneq B$ helps: a fixed-point-free automorphism of order 3 maps no five vertices onto itself but maps each of its 3-cycles onto itself, and with C one of them the search keeps only the base states of B above a chosen base state of C, choosing in each orbit the one with fewest base states above it. Three unit-margin searches of Appendix C kept 600 of 2,200 in this way.

Using the whole group. We take $H = G_R$, the common stabilizer of the representatives. For a single representative r the orbit–stabilizer theorem (Dummit and Foote 2004, Section 4.1, Proposition 2) gives $|\mathrm{Aut}(T)| = |\mathrm{Aut}(T) \cdot r| \cdot |G_r|$, so anchoring and the quotient between them use the whole group; with several representatives we choose them to keep G_R large, and the base among the unions of cycles of G_R or the sets containing one. Table F.1 shows the split: for the Paley tournaments the stabilizer of a representative pair is trivial, so the whole group goes to anchoring, and for $\Delta_1 - v_{10}$ it splits into orbits of 11 and a stabilizer of order 5; Figure F.1 summarizes which parts of Lemma 5.1 apply to a given tournament and base.

Table F.1: How the searches and certificates of Section 6 use the automorphism group. Anchoring uses the orbits of the representatives and the quotient their common stabilizer. In the $\Delta_1 - v_{10}$ search the representatives lie outside the base, so the count of base states kept reflects the quotient alone.

tournament	$\lvert\mathrm{Aut}(T)\rvert$	anchoring	quotient	base states kept
P_{19}, unit, certificate	171	ordered pairs	none	22,876 of 142,251
P_{23}, search	253	ordered pairs	none	2,591 of 8,031
P_{23}, certificate	253	ordered pairs	none	343,896 of 3,414,729
$\Delta_1 - v_{10}$, search	55	x_0, y_0 (orbits of 11)	order 5	1,611 of 8,031
$\Delta_1 - v_1$, $\Delta_1 - v_2$	5	none	order 5	1,611 of 8,031
$\Delta_{11} - v_2$, $-v_5$, $-v_{20}$	5	none	order 5	1,611 of 8,031
$\Delta_6 - v_5$, $\Delta_{28} - v_5$	1	none	none	8,031 of 8,031

F.3 The five steps on a small example

Given T and the margin regime, the search has five steps:

1. **Pick a size.** Choose the base size b, in practice 5 or 6.
2. **Pick a subtournament.** Choose a set B of b vertices. Only $T[B]$ matters for what follows, so the choice can and should be optimized quickly (Section 5.3), since the base-state count varies by more than two orders of magnitude across the 12 isomorphism classes at $b = 5$.
3. **Find its base states.** Enumerate the k-tuples of linear orders of B whose supports agree with $T[B]$, counted up to voter permutation by Lemma 5.1(d).
4. **Determine the usable symmetries.** Compute $\mathrm{Aut}(T)$ with `nauty` and its orbits, then impose the ordered-pair anchoring of Lemma 5.1(b), the vertex anchoring of Lemma 5.1(c), or neither, discarding the base states no anchored witness can use. If a subgroup H maps B, or a subset C of it, onto itself, also keep only the base states above one representative of each H-orbit (Lemma 5.1(e)), checking that H fixes the anchoring representatives (Figure F.1).
5. **Run to completion, in parallel across base states.** The verdict is positive as soon as any base state yields a witness. It is negative only when *every* live base state has been searched to exhaustion.

Figure F.2 shows all five steps on P_7 at $k = 3$. Steps 1–4 are fast; step 5 runs in parallel.

F.4 Runtimes for every computation

All runs used a single laptop (Apple M-series, 14 cores, 24 GB RAM, at most 11 cores at once) except the following, which used a compute cluster: the census of regular tournaments on 15 vertices, the self-converse census of order 13, the order-12 census of Appendix A, six of the 22-vertex sweeps of Section 6.5, part of the certificate for $\Delta_1 - v_{10}$, 16 of the 19 unrestricted sweeps of doubly regular tournaments on 23 vertices in Section 6.4, and most of the search of Appendix C.

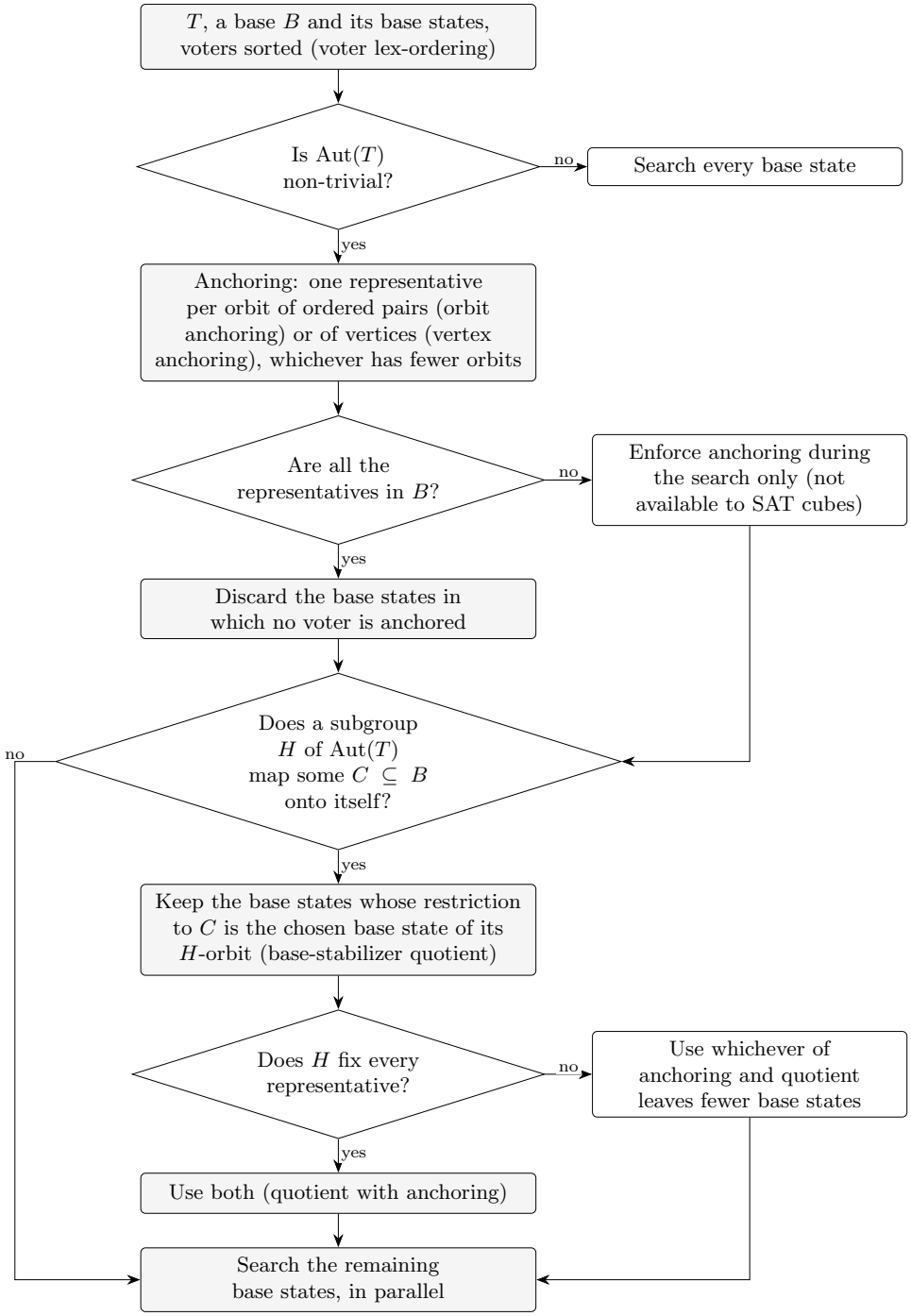


Figure F.1: Which symmetry reductions apply to a tournament T and a base B. Anchoring by Lemma 5.1(b) or (c) can discard base states only when every representative lies in B, since a base state reveals the orders on B alone; otherwise the search enforces it during insertion, and the SAT route cannot use it. The quotient of Lemma 5.1(e) combines with anchoring only when H fixes every representative.

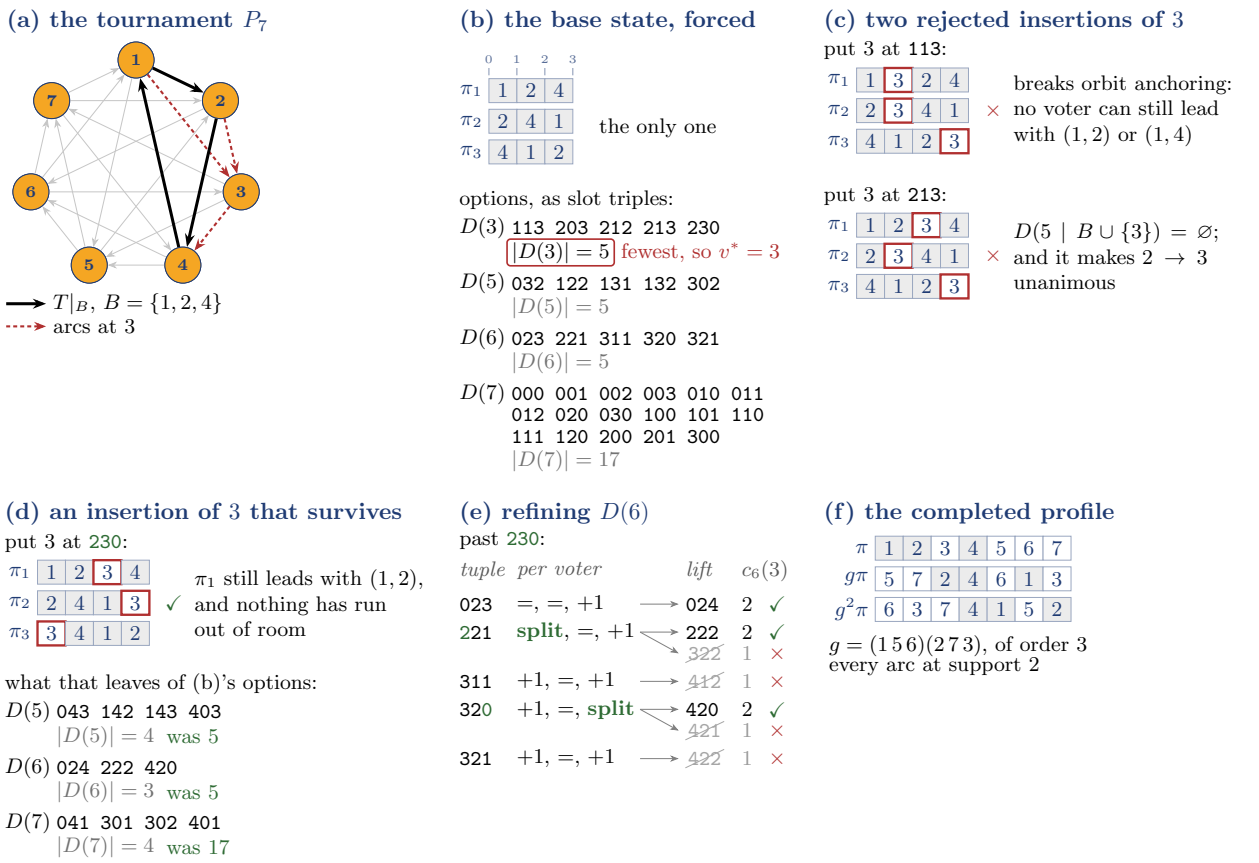


Figure F.2: Algorithm 1 on P_7 at $k = 3$, which is 3-inducible, so the run ends in a witness. A *slot triple* is one gap per voter, of the 64 available, and a vertex's domain is those triples its arcs to B admit. Here the base is $B = \mathrm{QR}$, the squares modulo 7, and Lemma 5.1(b) guarantees that at least one voter must rank 1 first and either 2 or 4 second. Ranking 4 second forces the other two voters to 241, the only order getting both $4 \to 1$ and $2 \to 4$ right, and $1 \to 2$ is then left with support 1; so the base state in (b) is the only one. **(a)** The base $T[B]$ on $B = \{1, 2, 4\}$ in bold; dashed, the arcs at the first vertex inserted. **(b)** Steps 2–4: the base state is forced, and MRV takes the smallest domain. **(c)** Two insertions the search rejects, one by Lemma 5.1(b) and one by leaving 5 no slot at all. **(d)** One that survives, and what it leaves of (b). **(e)** REFINE lifts each parent tuple past the slot just used (unchanged below it, shifted above it, split where the two coincide, which is what green marks), and the one new arc then keeps or kills each lift. **(f)** The witness, every arc at support exactly 2. Without Lemma 5.1(b) the search returns all 7 witnesses, a single orbit of $\mathrm{Aut}(P_7)$, of order 21, so the profile is unique up to symmetry, as Shepardson and Tovey (2009) report; π generates the other two under the g shown, the map $x \mapsto 2x + 4$ of $\mathbb{F}_7$.

Runtimes are in *core-hours*, the workers' running times summed. The cluster ran 2 to 5 times slower per unit of work, so we convert its runtimes to the laptop's scale as follows, unless marked as cluster core-hours.

The certificate for $\Delta_1 - v_{10}$ ran partly on each machine; on the 133 groups of five cubes solved on both, the cluster took on average 2.1 times as long (1.6 to 3.1 for individual groups), and we divide its cluster time by that average, so the converted total is approximate. The searches of $\Delta_1 - v_1$, $\Delta_1 - v_2$, $\Delta_{11} - v_5$ and $\Delta_{11} - v_{20}$ run a different program: on 355 base states of $\Delta_{11} - v_5$ searched on both machines, the cluster took 3.55 times as long in total, and we use that factor; we apply it also to the sweeps of $\Delta_6 - v_5$, $\Delta_{28} - v_5$ and 16 of the doubly regular tournaments on 23 vertices, for which no paired measurement exists. The self-converse census of order 13 ran partly on each machine, and there the cluster took 5.13 times as long per tournament. The censuses of orders 12 and 15 ran on the cluster alone; on random samples of 8 and 12 of their parts rerun on the laptop, the cluster took 3.85 and 4.28 times as long respectively. Table F.2 lists the result.

Table F.2: The runtime of every computation in this paper.

computation	margin	verdict	core-hours
P_{19}, certified refutation	unit	not inducible	25.1
P_{23}, certified refutation	unrestricted	not inducible	236.0
$\Delta_1 - v_{10}$, sweep (Lemma 5.1(e))	unrestricted	not inducible	96.9
$\Delta_1 - v_{10}$, certified refutation	unrestricted	not inducible	732
$\Delta_6 - v_5$, certified refutation	unrestricted	not inducible	1,292.7
$\Delta_6 - v_5$, sweep	unrestricted	not inducible	437.0
$\Delta_{28} - v_5$, sweep	unrestricted	not inducible	433.2
16 doubly regular on 23 vertices	unrestricted	not inducible	7,458.2
$\Delta_1 - v_1$, sweep (Lemma 5.1(e))	unrestricted	not inducible	136.8
$\Delta_1 - v_2$, sweep (Lemma 5.1(e))	unrestricted	not inducible	122.9
$\Delta_{11} - v_2$, sweep (Lemma 5.1(e))	unrestricted	not inducible	108.5
$\Delta_{11} - v_5$, sweep (Lemma 5.1(e))	unrestricted	not inducible	124.2
$\Delta_{11} - v_{20}$, sweep (Lemma 5.1(e))	unrestricted	not inducible	93.6
$P_{31} - v$, sweep	unrestricted	not inducible	239.5
P_{23}, one arc reversed	unrestricted	inducible	210.4
second doubly regular 19, all 57 arc reversals	unit	all inducible	3.59
regular on 15 vertices	unit	all inducible	726.1
self-converse on 13 vertices	unit	all inducible	294.2
all on 12 vertices	unit	all inducible	2,615.6

Table F.3 gives the four full Paley sweeps; P_{27}, built over $\mathbb{F}_{27}$, is not 5-inducible but does not bear on $N(5)$. On a fixed base decomposition the runtime falls as q grows, because a larger Paley tournament is more constrained and its search is refuted nearer the root, even though the work per node rises from 10.8 to 57.4 microseconds. Deleting a vertex, which shrinks the automorphism group, has the opposite effect ($P_{43} - v$ takes 185.5 core-hours), and on $P_{43} - v$ unit margin is about 90 times faster than plain majority, with 2,200 base states instead of 8,031 and the last voter forced.

Table F.3: The four full Paley sweeps.

q	live base states	core-hours	nodes	seconds per live base state
23	2,591	34.03	1.14×10^{10}	47.3
27	2,537	31.04	6.22×10^{9}	44.0
31	4,007	26.89	3.43×10^{9}	24.2
43	2,537	22.87	1.43×10^{9}	32.5

Appendix G. The obstructions on 22 vertices

Each obstruction of Section 6.5 is given as in McKay's (2026) catalogue: 231 bits, the upper triangle of its adjacency matrix row by row (1 when the u-th vertex beats the w-th, in the parent's labelling minus the deleted vertex), under the order of its automorphism group and any generators.

$\Delta_1 - v_1$ $\quad$ |Aut| = 5, generated by (2 9 11 7 5)(3 8 4 12 6)(13 22 15 16 20)(14 17 19 23 18)

```
11111000001111110000001100011110001111100100111000010110011111100110110010101
11011010101100110101011100101100110100110000101110010101101101010101100011100
10101011001011101101001100010110101011010101111000001101010001000101011110011
```

$\Delta_1 - v_2$ $\quad$ |Aut| = 5, generated by (1 19 12 23 8)(3 6 4 18 14)(5 16 15 13 7)(9 22 11 20 21)

```
11111111110000000000001100011110001111100100111000010110011111100110110010101
11011010101100110101011100101100110100110000101110010101101101010101100011100
10101011001011101101001100010110101011010101111000001101010001000101011110011
```

$\Delta_1 - v_{10}$ $|\mathrm{Aut}| = 55$, generated by (2 9 11 7 5)(3 8 4 12 6)(13 22 15 16 20)(14 17 19 23 18) and (1 18 4 14 23 19 12 6 8 17 3)(2 16 21 7 13 5 11 9 22 15 20)

```
11111111110000000000011111000011111100000011000111000111110010011000010110011
11110110110010101110101010110011010111100101100110001100001011101010110110100
10101011001011101101001100010110101011010101111000001101010001000101011110011
```

$\Delta_{11} - v_2$ $|\mathrm{Aut}| = 5$, generated by (1 21 12 22 9)(3 14 13 7 16)(4 17 18 15 6)(8 11 10 23 19)

```
11111111110000000000001110110001100001111001010110011101001110101100010111100
11101011001010011111011010111011000101011100111000101011001011010110001110101
11001001011100011101010001011110010011001010101000101010110010001011101110111
```

$\Delta_{11} - v_5$ $|\mathrm{Aut}| = 5$, generated by (1 21 12 22 9)(3 14 13 7 16)(4 17 18 15 6)(8 11 10 23 19)

```
11111111110000000000011111000001111110000001101100011000011100101011001110100 1
11101011001010011111011010111011000101011100111000101011001011010110001110101
11001001011100011101010001011110010011001010101000101010110010001011101110111
```

$\Delta_{11} - v_{20}$ $|\mathrm{Aut}| = 5$, generated by (1 21 12 22 9)(3 14 13 7 16)(4 17 18 15 6)(8 11 10 23 19)

```
11111111111000000000011111000001111110000011101100011000011100101011001110101
11010110001011100111010110010101111101101011101000101011100110001010110011101
01100011010111001000111000111101000101110010010010101100010010110100101101111
```

$\Delta_6 - v_5$ $|\mathrm{Aut}| = 1$

```
11111111110000000000011111000001111110000001100110100100111100110001000011101
11011100010110110111111111011110000101011001001011101010101011101001101010 1101
10110011001101001101011000110011011010101000111001001010100110010011100101110
```

$\Delta_{28} - v_5$ $|\mathrm{Aut}| = 1$

```
11111111110000000000011110000011111100000011101001000101011101010100001011101
11100110010110111011111111011110000011011010001110001110001010111010101011 0011
01101010110001101101011000110101011011010000111000100110000011010011100101110
```

$\Delta_1 - v_{10}$ **with the arc between** v_2 **and** v_3 **reversed** $|\mathrm{Aut}| = 1$

```
11111111110000000000011110000011111100000011000111000111110010011000010110011
11110110110010101110101010110011010111100101100110001100001011101010110110100
10101011001011101101001100010110101011010101111000001101010001000101011110011
```

Appendix H. Criticality at 21 vertices

Table H.1 reverses one arc per arc orbit, at unit margin, in the four vertex-transitive unit-margin obstructions on 21 vertices (Section 6.3): `U` if the result is still not unit-margin inducible, `S` if it is.

Table H.1: One arc reversal per arc orbit of the four vertex-transitive unit-margin obstructions on 21 vertices.

tournament	Aut	the ten orbits	obstructions
C_a	$\mathbb{Z}_{21}$	U U U U U U U U U U	10
C_b	$\mathbb{Z}_{21}$	U S U S S U U S S U	5
F_a, F_b	$\mathbb{Z}_7 \rtimes \mathbb{Z}_3$	S S S S S S S S S S	0